CHAPTER 9

Entertainment Applications

As we have seen in the previous chapters of this book, virtual reality has been put to use in a very wide array of applications. Scientists, businessmen, artists, and many others have utilized VR in ways that have helped them in their professions. The general public, however, is most likely to have been exposed to virtual reality in some area of entertainment. Whether it is via coverage of VR in the media, portrayed in some science fiction movie, or experienced first hand as a game or some other form of entertainment, the most likely avenue for the public to encounter virtual reality is in the realm of entertainment. While most people won't ever witness a physician practice a surgical procedure, they may have the opportunity to immerse themselves in a head-mounted display in an arcade and fight the evil forces.

9.1 AREAS OF APPLICATION

Virtual reality has spanned the entire continuum of the entertainment production life cycle from aiding in the creation of entertainment that is delivered in some other medium to being the end product as an entertainment product in its own right. A number of entertainment applications are illustrated in this chapter. The areas of application covered include the broad categories of *virtual reality experiences* and *entertainment production*. Each of these areas has multiple subcategories of note. Some of these subcategories include games, interactive fiction, creative expression, virtual set design and use, and virtual cameras used to choreograph computer graphics animations created for film or video delivery.

9.1.1 Games

Computer games have permeated society and become significant in the world's palette of entertainment choices. For the most part, computer games are delivered in arcade kiosks, at theme parks, on a traditional computer monitor, or on a television screen. All of these delivery mechanisms have a limited scope of interaction, and field of view/field-of-regard. In general, the participant is physically larger than other characters (whether the characters are computer generated or representations of other living people participating in the game, and there is no commonly used method to allow the participants to interact with others at life-size scale. Additionally, in traditional computer games, for the most part the participant's body is not integrated in the experience other than moving a joystick, pressing a button, or manipulating some sort of prop like a plastic weapon.

Some arcade games have integrated the participant's body to a greater degree by allowing them to stand on a surfboard or some other platform and use their body as an input device to the game. However, in all these scenarios, the participant must always physically face the screen, and the participant's head position is not integrated into the displayed point of view. Very few traditional computer games utilize stereoscopic imagery to create a three-dimensional effect.

Extending games into the medium of virtual reality offers many new opportunities for game play. VR offers the opportunity for true immersion into the game with up to a full field-of-regard, stereoscopic imagery, and natural user interfaces such as allowing the participant to physically jump rather than press a button on the controller to make their character jump. In general, traditional computer games operate from a second-person point of view (the participant is controlling their character that they see on the screen via some controller device). VR offers the opportunity for the game player to jump into the game and participate from the first-person point of view. To date, most VR applications operate from the first-person point of view with the notable exception of the *MANDALA* series of second-person point-of-view games.

9.1.2 Interactive fiction

Interactive fiction has existed for decades in media ranging from print through computer-based interactive fiction whether the experience is text on the computer or fully interactive computer graphics. The idea behind any interactive fiction experience is that there is a story line that the participant can alter based on choices they make throughout the experience. In a printed (text) interactive fiction, the person may read a page, and then go to one page if they choose to answer a dilemma one way, or a different page if they answer it another way. On the computer, the branching at decision points may be less obvious to the participant, but the idea is generally the same. With interactive fiction, there is the potential for each participant to have a different experience with the activity, and also for any given participant to have a different experience the next time they play it. One early, inspirational example of interactive fiction in the medium of virtual reality is *The Thing Growing*, created by Josephine Anstey and Dave Pape. In *The Thing Growing*, the participant interacts with computer-generated characters, generally in a *CAVE* display, though the creators have also made the application available for lower-cost displays. Throughout the story line, the participant is presented with an ethical dilemma and the outcome is based on the choices and actions that the participant makes.

Throughout the experience, there is a character that suggests things for the participant to do. The participant can choose to do them, or to ignore the character and do something else. In general in this application, if they ignore the character, the character gets more adamant and demanding of the participant. The experience culminates in a dilemma of whether to kill or not to kill with differing consequences. An interesting aspect of this application is that it was created especially for the medium of virtual reality (as opposed to translating an extant interactive fiction into VR) and exploits various aspects of the VR environment. For example, the main character instructs the participant to dance. The system utilizes the three tracking sensors (head, left hand, and right hand) to determine whether the participant did the dance correctly or not, with appropriate consequences.

9.1.3 Creative expression

Virtual reality offers the opportunity to freely create, sketch, and imagine in a three-dimensional world. In the simplest sense, one can use a three-dimensional drawing program to free oneself of the constraints of the two-dimensional world of paper or computer monitor. VR applications like *BLUI* allow one to create flowing, colored shapes in a darkened virtual world. In this book, *creative expression* applications are primarily covered in the chapter on art, but they bear a mention in this chapter because many people find the activity of using these tools find them very enjoyable. In addition to creating visual worlds, sound worlds and other senses can be represented in the virtual world as a means for creative expression.

9.1.4 Entertainment production

In addition to virtual reality applications being used by participants as entertainment in its own right, virtual reality has also been used as a tool for creating media (whether the media is virtual reality, music, film, or others) that is then experienced by the participant. For example, virtual reality applications can be used as a tool for aiding in the placement of three-dimensional sounds for playback in a standard surround-sound environment. This section focuses on applications of virtual reality related to the creation and use of virtual sets and for aiding in creating camera paths for computer graphics animation and preplanning camera moves for live-action film.

This chapter merely scratches the surface of some foundational application of VR in the entertainment arena. Many lessons can be learned from studying these applications and thus aid in the creation of new, remarkable applications both for entertainment and the creation of entertainment.

CASE STUDY 9.1: DISNEY'S *ALADDIN ADVENTURE*

When it comes to using technology to tell a story, few would argue that the Walt Disney Company does it well. Early in the development of VR technology and applications, the Disney group sought to explore how VR could be used for entertainment. One of their early efforts was to adapt the story from the box office smash *Aladdin* to an immersive theme park attraction.

In order to fully exploit the capabilities of VR, the Disney Imagineering group had to develop new hardware, software, and even a new computer language that allowed the creative team to directly build the application without requiring the intervention of technical personnel to implement every artistic choice or change.

The *Aladdin VR experience* has undergone many changes and developments over its life time, including a change from an exploratory interactive fiction world to a mission-oriented game. The Disney team had ample opportunity to observe and survey thousands of theme park guests regarding their interaction with this new form of entertainment, and fold the lessons they learned into new versions of the experience.

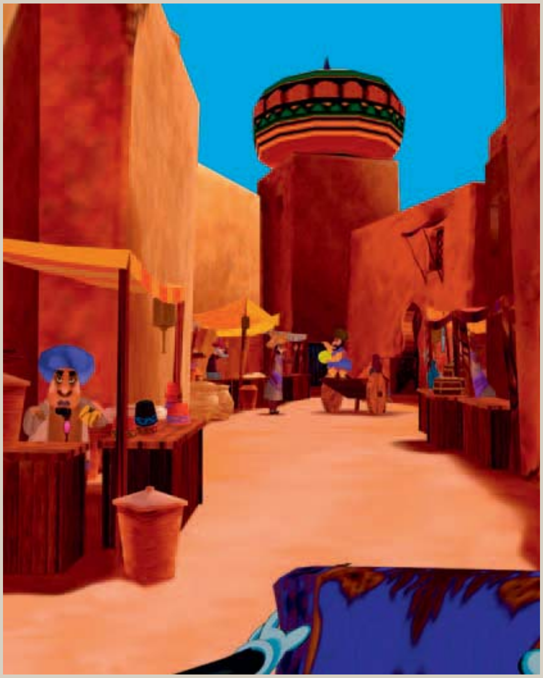

FIGURE 9-1
Disney's Aladdin Adventure *virtual reality experience allows guests to enter the city of Agrabah, the setting for the Disney movie* Aladdin.

Introduction

Walt Disney's *Aladdin Adventure* VR attraction demonstrates several techniques of entertainment and storytelling within an interactive immersive medium. Designed as an experiment to study the storytelling possibilities of virtual reality in a public venue, the *Aladdin* attraction went through many changes as surveys were tabulated and the audience experience was analyzed. The initial experiment was conducted at Disney's Epcot theme park near Orlando, Florida, from July 1994 through 1995. After these tests, a game-style narrative was written, and the revised experience ran at the Disneyland Starcade for several months in 1996. Following this it was removed from the park. VR then re-emerged in a Disney venue at the large family-oriented DisneyQuest arcades. The new venue opened in 1998 with new VR experiences based on the *Aladdin* experiments, along with Disney's *Hercules* feature-length animation, an experience that puts the guest inside a comic book, and another allowing a raft of guests to paddle down a prehistoric river.

As a leader in using new technology, Disney wanted to explore how this new (technology-based) medium could be used to tell a story for a larger location-based entertainment (LBE) audience. During the preliminary period of their research the feature movie *Aladdin* was still in early design stages. This film was ultimately selected as the setting for this experimental storytelling experience.

In many respects, virtual reality is just the next step in a progression of avenues used by theme park and other large LBE venues to tell a story to a group audience. Current methods include thrill rides, shows, game arcades, large-format motion pictures (sometimes in 3D or more), etc.[1]

[1] In addition to stereo images, some large format venues might include "gimmicks" such as light water spray, etc., corresponding to what is occurring in the story—e.g., MuppetVision*4D.

Interactive experiences abound in LBE parks, with bumper cars, carnival games, and even with the park itself often being an immersive experience (e.g., walk-around characters and parades down "Main Street"). Previously, the only computer-generated interactive experiences were the video games in the arcade. Immersive interactive LBE venues were only on the horizon when this work began, and the Walt Disney Imagineering team has gone a long way in exploring this new medium.

Application Description

The VR experience is a computer-generated world with scenery and characters similar to those in the Disney feature film. "Guests" are given a task, or mission that fits the theme. They are then given a fixed amount of time to fly through the world of Agrabah and complete their mission. To accomplish the assigned task, the participant must interact with the characters, and navigate through several locations in the virtual world while riding on the familiar magic carpet.

In the EPCOT version (1994–1995), a character in the world, the parrot Iago, gave brief instructions, including the task of finding the thief with half of the sought after scarab. In the 1996 version at the Disneyland Starcade, a park employee informed the participant of their mission, and often gave other verbal hints during the session.

In the Starcade version, the task set forth before the guest was to explore Agrabah and collect the five pieces of the magic lamp within 5 minutes. The pieces were the base, the body, the spout, the handle, and the lid. The lid was held by Jafar in the Cave of Wonders. Once four of the pieces were collected, the guest had to fly to the Cave of Wonders, and defeat Jafar by grabbing the fifth piece of the lamp, thus releasing the Genie.

DESIGN DECISION

In fact, there were more than five lamp pieces throughout the virtual world, and the participant only had to grab any four pieces. If a redundant piece was found, it would simply be added as one of the other remaining pieces. Once four pieces were found, the extra pieces would disappear from the game, thus still requiring the final confrontation with Jafar.

Giving people a quicker way to find the required four pieces was necessary because otherwise it would have been too difficult for people to accomplish the mission in the span of 5 minutes. As a side benefit, when players found lamps in a new location, it gave the impression that the experience changed between plays.

Agrabah, the virtual world in which the guest is placed, is set and populated similarly to its counterpart in the feature film. The characters in the world portray the same look and movement. The locations had the same look and were accompanied by a thematically similar musical score. Many parts of the world were so faithful to the world of the movie, that guests could use knowledge of the location of secret passages, etc., gained from viewing the movie to their advantage in the VR attraction.

Of course, one of the considerations faced when adapting a virtual world from a linear narrative medium such as a film to an explorative medium like VR is that in the linear version, many details are not entirely fleshed out. For example, only the main entrance to the throne room is shown, but since it would be logical for side doors to also exist, they were added in the VR experience. This also benefits the VR experience creators by allowing them to branch the story based on the participants' actions. Another example would be that the complete route from the palace to the Cave of Wonders is not shown in the movie. The action simply jumps from one location to the other. However, because people bring a sense of diegesis to the movie, they have a sense that they know the complete world of Agrabah, even though they were only exposed to portions of it during the film. The fact that some specific pieces of information, such as the location of the secret entrance to Jafar's laboratory, were consistent between the animated and VR virtual worlds, the perception that the two are one and the same is enhanced.

Content is the key to any mediated experience, and is more important than the technology used to deliver it. With a background in bringing worlds to life using many different media, the Disney team was very familiar with the requirement of having an interesting world, and working with the possibilities and constraints of the chosen medium. One aspect of this project was to experiment with different styles of content and forms of interaction.

In a virtual world explored via an interactive medium such as VR, there are many ways to permit the participant to explore the world. The Disney team tried a range of "navigational narrative" styles during the implementation of the various incarnations of the *Aladdin* experiences.

On the noninteractive end of the range is the ride-along–style navigation. In this method the guest is taken along a predetermined path as though they were on a rail in a mechanical theme park ride. In another method, the guest is fooled into thinking they were interacting with the world, but is actually being led through the world. The other end of the range was to allow the participant the complete freedom to do whatever and go wherever they choose in the virtual world.

DESIGN CHOICE

In the public exhibition version of the experience, the design team chose to put the "guest" into a world with a plethora of interesting activity taking place. This gave the rider the ability to explore, making choices that lead them down different branches of the "story." However, to keep the experience moving forward, timed events such as collapsing walls or earthquakes would prod along users who hadn't made sufficient progress. As in any storytelling medium, one gets to a certain point where a crisis is needed to move the story forward.

In an environment where any activity can be done in any order, an enormous range of sequences of actions are possible (a situation referred to as "combinatorial explosion" in computer science). This freedom increases the computational requirements for the system and makes it difficult to keep the storyline tractable and intact. It also increases the time required to design, implement, and test the many possible scenarios.[2]

DESIGN CHOICE

To handle the problem of a world with explosive possibilities, the amount of complexity of the world at any particular moment must be limited. The creation of specific *scenes* that provide a specific setting in the world helped the designers manage this problem. As users enter a new scene, they can only interact with the objects and characters within the scene. After interacting for a while, guests will either choose, or be forced, to move to another setting or scene. By exiting a scene, all events local to it are brought to a point of closure. The use of scenes also helps control the amount of branching that can occur during the experience. Liberal use of the isolated scene technique helped keep the job of creating the experience manageable.

Representation of the Virtual World

The content of the *Aladdin* virtual world includes the city of Agrabah and the surrounding desert. It also includes the objects that can be acquired and the characters with whom the guests can interact. The general style of representation is cartoonish, as it was in the animated film.

Internally, the cartoon-like world is stored as a polygonal database. The division of the narrative into scenes helps to limit the computational requirements needed during any phase of the experience. Character motion was created and stored as an animated sequence of positions. A specialized programming language was developed (Story Animation Language—SAL) to allow the narrative to be specified by the storytellers themselves.

Visually, the virtual world of *Aladdin* is represented in a style that closely resembles the style of the feature film from which it was derived. The primary difference is that in the VR experience, the world and characters were modeled and animated in three dimensions. Buildings and characters were painted by animation artists in a computer graphics environment.

The painted texture maps incorporated the lighting effects of the scene. This technique helps achieve the desired look as well as reduce the computation requirements of rendering. There is no need to perform complex lighting operations.

[2] Interactive fiction and DVD/CD-ROM games face this same difficulty.

FIGURE 9-2

The guest's hands are represented as an avatar that appears as white gloved hands holding a magic carpet. The guest navigates through the adventure by riding a virtual magic carpet.

A minimal avatar of the guests represents them as a pair of (white gloved) hands clutching the front edge of a flying carpet. The flying carpet is the virtual vehicle the participant rides and steers though the world. Even this limited avatar helps to ground the user into the virtual world.

Aural representation played a key role in the *Aladdin VR experience*. The *Aladdin* team felt that audio is an extremely important element of a VR application. Composer and sound designer Aarin Richard was brought to the team to work on this part of the experience and to research how 3D audio could be used to enhance the experience. He and producer Jon Snoddy discovered that different types of audio sources could be mixed without breaking the illusion, and this was used to create a more engaging sonic environment.

As with a feature film, the sound in the virtual world of *Aladdin* is a combination of many sources. General ambient sounds, as well as event- and location-driven sounds, were a major part of the world of the *Aladdin* "ride." Sounds included crowd noises in the market place, rumblings of an earthquake, and the voices of animated characters speaking to the participant. Event-driven (marker) sounds were used to indicate location or emotion in a manner similar to mood music in motion pictures. Each location had a specific introductory fanfare and ambient soundtrack.

One of the initial concerns with the design team was how difficult it would be to combine sound types. They thought it might be more complicated than with films, because not only did it need to be done in real time, but they had to combine different sound types. For example, collision sounds might be monaural, ambient music might be stereophonic, crowd noises might be user-grounded 3D recorded sounds (i.e., the sounds move when the user moves), and character voices might be presented as world-grounded spatialized 3D sounds (i.e., the sounds are stationary with respect to the virtual world).

VR DISCOVERY

In experimenting with the audio content for *Aladdin,* sound designer/composer Aarin Richard discovered that different types of sound could easily be mixed together without causing the listener to be disoriented. For example, while a spatialized crowd sound plays without relation to head movement, the voice of a specific character talking to the guest is rendered relative to the orientation of the participant's head. The combination of all these sounds resulted in a satisfactory sonic experience.

By associating audio cues with objects in the world, the team found that the level of immersion of the experience increased. This occurs because even though an object leaves the user's view, spatialized audio cues continue to indicate that the object continues to exist in its proper place in the world. When objects have no localized audio associated with them, users often move their head to follow or search for the objects, trying to keep them in view and verify their permanence.

VR DISCOVERY

User perception of object permanence is increased when objects have an aural representation that is rendered to emanate from the location of the object.

In addition to places and objects providing unique sounds, certain events also generate their own sonic representation. Collisions between the flying carpet and objects in the world were one type of event presented sonically. The exact sound heard depends on with what the guest collided. A "palette" of sounds was created for different objects and characters. Each object and character had multiple sounds associated with it. These sounds were cycled through to prevent annoying repetition of a sound that could ruin the experience of a guest who was having trouble steering away from an object in their way.

For collisions, the *Aladdin* authors experimented with different styles of sound representations. After experimenting with realistic sounds, they tried and settled with cartoon-like sounds.

DESIGN CHOICE

The experience authors chose to use the cartoon-like sounds to represent guest collisions with objects in the world. They felt that this worked better than realistic sounds, both because it matched the cartoon-like visuals of the world, but more importantly, because these sounds are more amusing. Since collisions occur when the user is having difficulty flying, hearing the light-hearted sounds is more likely to lower the amount of frustration that they are feeling.

Three types of haptic feedback were part of the complete trial experience. A breeze to indicate movement through the world was generated by a fan, a heat lamp was used to indicate when the guest flew near lava, and the physical mechanism that the participant holds onto to control the flight.

The fan and heat lamp methods of providing breeze and thermal feedback to the user were not successful. Users reported that they did not even notice the effects, despite tests with even larger fans. The reason these methods didn't work was due to the fact that the HMD's worn by the users covered the part of the body that would have been most likely to notice, the head. As HMD's shrink in size, devices such as fans and heat lamps may become more feasible.

One method of haptic feedback that was found to have a positive effect on the experience was the physical mock-carpet device used to provide travel control to the system. Although the carpet prop does not actively feed back forces or resistance to the user, physically holding the prop provides sufficient proprioceptic feedback to close the interface loop with the user. This simple, "passive haptic" feedback makes controlling the carpet easier.

Providing a prop that is consistent with what the participant sees and hears has the additional benefit of helping make the entire world seem more real. We refer to this method of providing passive physical objects that mimic the expected haptic feedback the *transference of object permanence*, that is, the physical nature of what is touched makes the rest of the world seem more tangible.

The motorcycle-style platform that the guests mount to "ride the experience" was also capable of providing vestibular feedback via a motion base that could raise and lower, and tilt side to side. Participants reported an increased sense of immersion corresponding to the magnitude of motion feedback, but reports of discomfort due to nausea also increased.

DESIGN CHOICE

Vestibular feedback was disabled from the public version of the experience due to the higher level of reported nausea. The *Aladdin* team felt that the level of involvement provided by the head-mounted display was enough that the added benefit of vestibular feedback was comparatively insignificant, and many participants did not seem to notice the lack of motion.

However, an easy method of providing motion feedback was chosen by using "tactile seats" that provide a vibration that the rider feels as a "thump."

As with most VR applications, the olfactory and taste senses were not implemented. In addition to the limited availability of hardware, the health hazards concerned with an experience designed for use in public venues make synthetic taste and smell even more impractical.

Interaction with the Virtual World

Originally, the *Aladdin* experience was designed in the genre of *interactive story*. That story was presented to the user as a world to explore, and characters and objects to interact with, where each guest was part of their own story. However, with the specific task to accomplish added to the experience, it becomes more of a *mission game* experience. This style of navigation/interaction is the platform based *vehicle pilot-through* form. In this style, all the interaction with the virtual world is done by navigating through it and flying close to characters and objects.

Since the application was created to explore the narrative possibilities of the new medium, the design team was able to make many observations about the medium. One such observation is that the author no longer has complete control over the scenic and timing elements of the experience. Control is lost as the interactivity is increased from virtually none in a rail-style ride-along experience to almost unbounded interactivity possible in a typical VR walkthrough.

A corollary observation (the flip side) is that given no direction on what to do in a world, the creators noticed that in their initial, limited world, participants remained engaged in the experience for up to 2 minutes, and shortly after that asked for guidance in what they should be doing. The narrative needed to be more than simply go out and explore a new world. The size of the world and number of ways to interact with the world are probably a large factor in how much time people will remain engaged without direction.

VR DISCOVERY

For missions longer than the time required to adequately venture through the entire world (e.g., 2 minutes in the EPCOT version), participants expected to be given a goal.

The guests in the world of *Aladdin* are therefore given a goal to make the experience interesting. In general, people want to be directed. They want a mission to accomplish. This desire to be directed might be heightened by the fact they are in a theme park venue that customarily provides packaged entertainment to the guest.

In an LBE, the goal of the experience is often introduced during a nonimmersive, "preshow" presentation (e.g., a video segment). This can be used to give both a history of the world leading up to the guest's mission, but also to give instruction on how to travel and navigate through the virtual world. In a high throughput environment, such as a Disney theme park, the immersive experience of one group can overlap with the preshow of the next group.

VR DISCOVERY

Both the throughput and the overall experience can be improved if a pre-immersion "background story" is used to get the user "up to speed" on the events/culture/situation of an LBE virtual world [Pausch et al. 1996]. However, in arcade venues, which tend to be more "self-serve," kiosks improving throughput with this method is more difficult since the instructive preshow would likely still require the use of the VR system. On the other hand, many people learn by watching the play of previous players.

As with the natural progression of creating interesting content for any medium, the narrative went through many changes as the project was refined. Major changes in the story were made as it progressed from its original concept, to what appeared in the experimental version at EPCOT, and then to what ultimately became an arcade-style game at Disneyland.

The original idea was to present each guest with a 5-minute personalized story that takes place in Agrabah, the world of *Aladdin*. As the main character in the story, the guest would "live" the narrative based on the characters, locations, and other elements of the feature film. The story was not intended to be gamelike, but random events would move the narrative down different paths, giving a different experience to each guest.

DESIGN CONSTRAINT

Difficulties abound when creating an interesting and unique narrative within a world as complicated as that of *Aladdin*. This condition is exacerbated when creating an experience for the conventional LBE timeframe of 5 minutes. So, the narrative was designed to move the user from event to event, with fairly tight control over the path of the participant.

The experimental version of the attraction that ran at EPCOT had a few gamelike elements, but mostly consisted of the user exploring the city and interacting with (i.e., listening to) some of the characters. In fact, the goal of this version was simply to locate a specific character (a thief with the other half of a scarab) by using hints provided by characters met along the way. Once the thief is thwarted, he gives the player the missing piece of the scarab and the experience ends.

Additional game elements were added to the version that ran in the Disneyland Starcade during 1996. Expert game designer Jesse Schell created a mission for the guest to accomplish during the 5 minutes of each experience. That task was to explore Agrabah and the surrounding locations searching for four of the five pieces of the lamp. They were then to fly to the Cave of Wonders for a final battle with Jafar. Upon entering the cave, Jafar would transform himself into a giant snake with the fifth and final piece on his tongue. To defeat Jafar, the participant had to grab the fifth piece of the lamp. Doing so would cause the Genie to be released from the lamp, Jafar to be defeated, and all to be well in Agrabah.

VR DISCOVERY

One interesting feature of making the guest search for something is the ability of the game designer to force the participant to consider the new capabilities of the VR medium. At least one piece of the lamp required them to physically look up to see it, something not possible in video games displayed on a standard monitor.

Of course, as with most arcade-style games, it is difficult if not impossible to accomplish this on the first attempt. Since there was no way to accrue time above the allotted 5 minutes, any time spent exploring side locations or listening to the world's inhabitants came out of valuable time needed to collect the pieces and defeat Jafar.

To be successful, the guest would first have to explore the world a few times, gathering the information and advice from the characters and locating the lamp pieces. Then, to win, they would re-enter the world, fly directly to the pieces, then to the Cave of Wonders and Jafar, ignoring everything else in the world. In fact, given the time constraint, some first-time players chose to ignore characters who were speaking to them (during which time they are prevented from flying away) and begin to look in other directions, focusing on the task of locating the lamp pieces.

As the guest explored the world, the mood of the experience could also be manipulated. The *Aladdin* Imagineering team implemented and tested techniques to create a mood for a scene. Specific music or lighting effects can be chosen based on the guest's current situation. A turning point in the experience (e.g., finding the palace) might cause a general shift of music and/or lighting, and therefore the mood of the guest. In more complicated narratives, the mood effects for a particular scene may be based on previous events in the guest's experience.

In experimenting with several narration techniques, the application authors felt that they had found several good ways to compose scenes designed to draw the guests' attention to specific details. Once the guest is looking toward a character, that character can point to objects or bring an object into view by walking [Pausch et al.].

Similarly, techniques to virtually move the carpet (with the guest aboard) were sometimes tried to help move the narrative along. For example, the carpet might be dragged by a character to move the guest to a specific location. The experience developers experimented with narrative techniques such as preventing flight backtracking by closing doors behind the guest, or by forcing the carpet to continue moving forward as a means to keep the story moving [Pausch et al.].

DESIGN DECISION

One method of travel that the experience creators experimented with was the "tow-rope" method, in which the carpet seems as though it were being pulled at the end of a long rope. In this method of travel, users can still move left and right, but only for a limited distance, and they remain pulled by the "rope." To allow more freedom to the participants, the creators later chose a less restraining method that used the looser concept of attractors to draw the carpet in certain directions. Guests were free to turn around and thus break the attraction.

General navigation through a three-dimensional space with 6-DOF control is very difficult. The majority of the population has little previous real-world experience with such navigational techniques. There are techniques that seem intuitive to some of the population, but might be counterintuitive for others (e.g., pilots vs. nonpilots).

DESIGN CHOICE

The *Aladdin* experience uses a piloting metaphor to transport the guest through the virtual world. As in the world of the film, flying is (seemingly) accomplished by piloting a magic carpet. Users grasp what feels like the edge of the carpet, and can tilt it from side to side, pitch it up or down, and push or pull it to control turning, climbing, and accelerating.

Since most of the interaction with the world is accomplished by flying to characters, objects, and specific locations, the importance of a good travel interface that keeps the user from floundering around or getting too lost is very high. To safeguard against these eventualities, user movements are constantly analyzed, and if they are having trouble or are far off course, the application will gently move them along in a more fruitful direction.

DESIGN DECISION

To enhance the realism of the virtual world, the creators chose to prevent guests from flying through walls. The result of this decision is that guests could become stuck should they fly into a tight corner.

The creators experimented with allowing backward flight to alleviate the frustration of becoming stuck. In the Disneyland version of the experience, backward flight was allowed, although the story strongly encouraged the guest to move forward toward some goal. For situations where reverse travel was not allowed, or the user was unable to successfully get out of a corner, the system would slowly redirect the flying carpet each time it would bounce off the wall, eventually freeing the guest from their predicament.

Gaining altitude required flying at higher speeds. This demanded some skill from the pilot in situations such as flying down the crowded streets of Agrabah. Once successful, however, the guest was treated to a breathtaking flight above the city.

Computer-controlled agents are a key ingredient in the world of the *Aladdin VR experience*. Most of the interactions with the world center around characters and the information they provide the guest.

The choreographed characters are designed to draw the attention of the participant. Once they are within close proximity, the character branches into a communication mode and relays information to the guest. The guest then chooses what course of action to follow (that is, where to fly) next.

In the virtual world of *Aladdin,* interactive characters served different roles. Some of the characters provided valuable

FIGURE 9-3
Guests are able to learn information about the world by interacting with computer-controlled agents that appear as various characters in the world.

information required to accomplish the mission. Other characters seemed to serve more to fill the world with interesting activities. In one version, Iago the parrot served as the instructor (a kind of narrator) at the beginning, and could fly along with the carpet giving hints. Jafar, of course, served the role of final antagonist that had to be defeated to *win the game*.[3]

Characters that played a crucial role in accomplishing the task were distinguished by the fact that they literally held the attention of the guest, by preventing them from flying away, until they made their statement. Characters with less important things to say would still interact with the guest within their role, such as street vendors trying to sell you something. These "filler" characters may not have advanced the story, but they did add dimensionality to the world.

VR DISCOVERY

The implementers found that using eye contact between the characters and the participant made the experience more engaging.

[3] Interestingly, the magic carpet itself doesn't seem to be an interactive character, as opposed to the movie. It might have been intriguing to have the carpet point to locations or be reluctant to go in certain directions during the course of the experience.

The movement and dialog of the agent-characters was scripted. Similar to standard animation practice, dialog was first recorded and then movement was formulated to match. This is typical of how animations are created.

VR DISCOVERY

In their initial experiments, body movements of the characters were created using a motion capture, performance animation system. This is also how motion is captured in many modern video games (especially sport games).

However, motion capture data is noisy and requires extensive cleaning and filtering. Because the Disney team happened to have expert animators available, they found that they could produce movements more quickly and more to their specification using keyframe animation techniques.

After some initial experimentation with multiple-guest experiences, the *Aladdin* team decided not to include this feature with the first full release in the Disneyland Starcade. They felt it would be too difficult to create a situation that enables multiple participants to collaborate on something worthwhile during the 5 minutes or so that they would be engaged in the experience.

However, 2 years later, DisneyQuest, a new customer that was more like small theme parks rather than the Starcade

arcade venue in Disneyland, requested a competitive/cooperative game. Also, by this point a single computer was fast enough to generate images for four players acting independently.

A multiple-guest experience introduces other difficulties in creating the application. For instance, if the music and lighting were adjusted for each guest, then it might be difficult to determine what effects would be experienced by the group, though something like the multiple-perspective method used by *CALVIN* (A VR application designed by the Electronics Visualization Laboratory at the University of Chicago at Illinois) could be used. In that system, each person sees the world in a unique way, even if they are in the same virtual space. Thus it would be possible for each guest to have their own theme melody that is played when other guests encounter them. This melody could be considered the audio portion of their avatar.

The physics of this virtual world are very simple. The physics consists of the flight dynamics for the guest to pilot the carpet through the world, along with other flight manipulations such as attractors to locations/characters and recovery of "stuck" guests. All other actions that occur in the world are specifically choreographed, and not the result of a simulation (cartoon-style or otherwise).

DESIGN CHOICE

After analyzing the problem of developing a world physics simulation, the designers came to the conclusion that there are too many things that work against accurately simulated physics. For example, some difficulties included only partially understood theories on how things work, and even if fully understood, the computational requirements to calculate the simulation in real time. Thus they decided they would take complete control and choreograph everything. They'd have to "cheat."

By making this choice, they can more easily create a world that is *fantastic* (i.e., in which "fantastic" things can happen, versus the mundane). Animators are also often capable of making events look "better than real." In other words, simulated falling blocks might appear fake to the participant, but an animated sequence can be made to look "right," with the added benefit that the result might better fit the narrative than a random placement of blocks.

Another strike against using simulated physics is that if it wasn't a complete simulation that included every detail, such as swaying trees, etc., then the world will look fake and cartoonish anyway. Therefore doing a partial simulation of the world doesn't do any good, and a full simulation isn't possible, because the computational requirements are prohibitive.

The participant does control their path through the world, thus a magic carpet flight simulation algorithm was necessary. A simple-but-flexible flight control system was developed that would enable most of the general population to figure out how to direct the flight of the carpet. Steering was accomplished by maneuvering the carpet control in three ways. The unit could be turned left or right to control the yaw. Pitch was controlled by tilting the unit up and down. Finally, speed was adjusted by moving the "carpet" closer to or farther from the chest. In 2D movement, this is similar to how one controls a horse. Adding flight results in a user interface that is not from any real-world vehicle, although people are able to fly reasonably well [Pausch et al.].[4]

In later versions, the altitude control was changed. It was observed that people generally aren't used to navigating in 3D space and thus got more confused and/or nauseated by the pitch controls. So the up/down tilting control was changed to simply translate the carpet and its rider up and down.

Venue

With the venue of Disney's theme parks in mind, the system was designed to be usable by as much of the guest population as possible. The experience had to be relatively easy to operate, it should cause no ill effects in most of the population, and it should fit most body and head sizes. There are also provisions for quickly removing the motorcycle-style mount to enable wheelchair access.[5]

[4] We were unable to locate any flying carpet or carpet pilots to compare with actual flying carpet controls.

[5] By giving the "ability" to fly to people who were not able to walk, the experience was often quite emotional for these guests and their families.

The presentation venue can also affect the practicality of the story and how it can be presented.

DESIGN CONSTRAINT

Disney's experience has shown that in ride-style LBE venues (vs. shows), experiences are economically feasible if they last 2 to 4 minutes.[6]

This is in stark contrast with a typical interactive narrative experience that people have access to on their personal computers, where a world may take weeks to fully explore (e.g., large interactive fiction, DVD/CD-ROM games, or many console video games).

Another major factor of the public venue is the concern of throughput. A general theme park ride or show needs to move about 2000 people through per hour. At EPCOT 100 people (for each 5-minute time slot) were given a demonstration that included four immersed participants selected from the audience. Smaller venues such as LBE's like DisneyQuest have lower throughput requirements, as do arcades. In an arcade, where the guests pay extra to experience the arcade machines, the constraint is merely that each game last about 5 minutes.

Some specific design choices were made for the EPCOT evaluation version of the experience. Because it was an experimental attraction, they were able to veer from conventional theme park "rides" and choose only a small subset of a group to actually experience the world *immersively*. From an audience of 100 people, 4 were picked to don an HMD and mount the four motorcycle-style platforms.

DESIGN CHOICE

For the experiment to address the needs of the entire theme park audience, a broad selection of people needed to be included, not just people with a natural affinity for new technology. Therefore, to ensure that a broad range of people were put through the experience, the designers specifically chose to select participants from the audience who did not seem to be overly interested in the technology. This was done by giving the criteria to the park guides who ran the attraction and selected who would be given the immersive experience.

VR System

Apart from the computer hardware, almost everything in the VR system for the *Aladdin* experience was designed and developed from scratch specifically for use in a theme park/LBE venue. In particular, the HMD and the vehicle (magic carpet) platform with motion base went through considerable design analysis of what was best suited for the venue. The HMD was designed for durability, health concerns, and usability by a high percentage of the

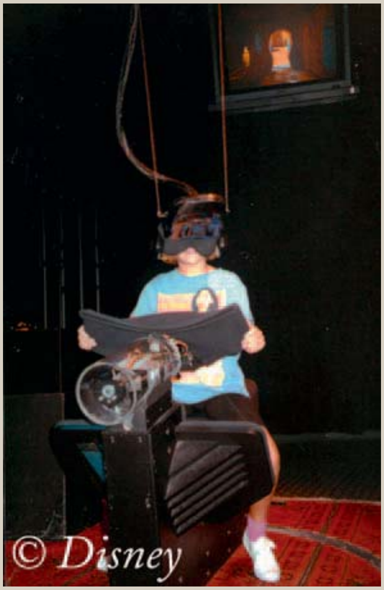

FIGURE 9-4
At Disney's EPCOT Center, some guests were allowed to mount a motorcycle-like platform and wear a head-mounted display while others watched.

[6]The overall design of the rides are created to make the experience seem to last longer, for example, the Star Tours attraction.

population. The carpet platform was designed to be comfortable for most people and to aid in the sensation that you were flying. Durability is very important because of the high throughput. One-in-a-million failures would be regular occurrences. [Pausch et al.]

The *Aladdin* team tested many display methodologies for presenting the virtual world to the participant. Based on that experience, they chose head-mounted visual displays and designed a model that fulfilled their design criteria.

SYSTEM CHOICE

Display Paradigm

In their initial assessment of the pros and cons of visual display technologies for VR, the Imagineering VR developers chose HMD over projection-style displays, such as the *CAVE*™. This was primarily based on cost and space limitations for a theme park venue. Theme park economics revolve around how many people can be moved through a certain sized area in some (short) amount of time. Putting a single person in a projection-based VR display means more space and more computer power would be needed for each guest than was feasible.

However, some other VR attractions produced by the VR studio for DisneyQuest did use a projection VR system. Specifically, the DisneyQuest experiences based on the themes of *Hercules* and later the *Pirates of the Caribbean* utilized projection displays.

DESIGN CHOICE

The HMD designed for the *Aladdin* experience was capable of providing a stereoscopic view of the world. However, the designers chose to use the two field-sequential CRT screens to provide a monoscopic image to the viewer. The HMD provided a 90-degree horizontal field of view (FOV), and 640 pixels horizontally by 480 vertically (i.e., NTSC resolution).

High-quality speakers were mounted near each ear for the aural display. To address sanitation concerns, separate helmet liners were attached to the head, to which the rest of the HMD was connected. These liners were then cleaned before re-use. The weight of the display is borne by cables that are suspended from above the participant.

This pixel count and FOV were chosen based on the design team's analysis of the minimal requirements for immersion into a virtual world. Specifically, they felt that an FOV less than 90 degrees resulted in a second-person view of the world, like peering through binoculars. Regarding resolution, they felt that using lower than 640 pixels simply made the image too fuzzy to see the world very well. At the time *Aladdin* was created, LCD displays could not meet these specifications, so CRT display technology was required. CRTs, however, add noticeable weight to the display, so a counter weighting system was devised to take some of the load off the guest.

Their HMD was designed to handle monoscopic images with bi-ocular, 3-inch CRT displays positioned to be in focus for a wide range of human heads. A single (wider) view of the world was rendered and presented (overlapping and cropped for each eye) giving the effect of a scene focused at infinity. Having noticed that many experienced VR practitioners who tested their system were often not sure whether it was a stereoscopic image, they felt that the potential problems with increased eye strain in improperly aligned displays made the benefits of stereoscopic images not worth the extra complication, and of course the additional (double) graphics computations.[7]

The primary inputs to the VR system were the carpet controls and the user's head orientation.

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[7]There are many ways in which our perceptual system can interpret the three-dimensionality of the world. Stereoscopic images are not absolutely required to do so. Depth cues are also provided by motion, occlusion, object size, etc. While head lateral motion is not available in the Aladdin experience, motion through the world is almost continuous, and is generally more helpful for scenes with many distant objects. The effects of head motion and stereopsis depth cues are diminished for distant objects and so are not helpful for many parts of the experience.

SYSTEM CHOICE

Haptic Display Paradigm

A motorcycle-shaped platform was designed for the guest to mount in order to operate (i.e., "ride") the magic carpet. The rider was able to sit comfortably on the seat and reach forward to hold onto what felt like the front of a flying carpet.

DESIGN CHOICE

Tracking

An electromagnetic tracking system was used to track the head motion of the user. Only the orientation information was used to render the scene.

The system designers felt that the noise of the electromagnetic tracking system was more noticeable in translational movement versus rotational movement. Deciding that a noisy channel of information is worse than having no information, they chose to only use the head's orientation, which is the minimum input required to make a head-based display function suitably.

SYSTEM CHOICE

Compute System

To provide maximum frame rate, each unit was supported by a Silicon Graphics Onyx workstation, equipped with three Reality Engine 2 graphics pipelines. Each workstation included four RM5 boards providing a total of 16 megabytes of texture memory—top-of-the-line graphics in 1994.

As the initial test version was developed for display at the EPCOT Center in 1994, a requirement of 60 visual frames per second (FPS) was set in order reduce the occurrence of nausea. To achieve this rate, each of the three graphics pipes rendered the scene at 20 FPS, with the resultant images chronologically interleaved, producing the overall rate of 60 FPS [Pausch et al.]. Another beneficial effect of this is that perceptually, the higher frame rate will make the world seem more real. While the lag at the onset of each image is no different than a noninterleaved method, this method reduces the average lag of the visuals.

For the DisneyQuest version, deployed in 1998, a single Silicon Graphics Infinite Reality graphics system was capable of providing a sufficient frame rate without requiring the added complication and cost of using multiple computers. In fact, the single IR system was able to render the monoscopic views for all four of the participants at the same time, resulting in a much more economic solution.

Application Implementation

Walt Disney Imagineering developed this application for the purpose of investigating the use of virtual reality in a theme park venue. While this effort resulted in a specific mission-game VR experience, the underlying storytelling environment allowed for great flexibility. During the experiment at EPCOT, this flexibility was exercised by changing the scenario from week to week to try different ideas with a large number of guests.

Having been introduced to the immersive power of VR by simple wireframe virtual worlds seen at the NASA Ames Research Center in 1989, attraction producer Jon Snoddy and others at Disney became interested in the possibilities of applying this emerging medium to the art of storytelling. They began to seriously investigate the possibilities of VR with an experimental project in collaboration with computer graphics flight simulation company Evans & Sutherland.

For the first experimental experience, a Disney movie that featured a character that does a lot of flying was chosen as the virtual world in which to tell such a story. As the movie title suggests, *The Rocketeer* gave a good excuse to give participants the ability to fly. However, as the movie did not do as well as expected in box office receipts, it was dropped in favor of another movie on the Disney horizon. *Aladdin* was in development in the Feature Animation department, and happened to also have a tie-in to flight with a friendly magic carpet. In addition to the promise of being based on a more successful motion picture, the choice of *Aladdin* offered the opportunity to make the computer-generated world have the same hand painted cel "look and feel" as the animated film. Though this task was by no means easy, it was less difficult than making a virtual world seem like the real world, as portrayed in *The Rocketeer*.

At the same time, a switch was made from specialized flight simulator image generators to workstation-class graphics machines. Graphics workstations from Silicon Graphics Inc. were not only showing signs of being able to handle the necessary graphics requirements but also held promise for transitioning to less expensive home and set-top units.

The cost of developing the *Aladdin* attraction was significant by VR industry standards, but it was far below the resource requirements of a typical theme park venue. A Disney attraction such as the "Indiana Jones Adventure" might take hundreds of people to create. The *Aladdin* ride was developed by a team of approximately 25 people over 2 to 3 years.

Development of the attraction, including story, hardware and software, was done entirely in-house by a diverse group of Disney employees. Because it was developed as a theme park operation, the creation of *Aladdin* was led by producers whose job was in the design and construction of theme park attractions. This team was led by Jon Snoddy, then director of Imagineering's efforts to include new technologies into the Disney theme parks.

There were many personnel who provided key ideas and support of the program. In addition to producer Jon Snoddy. Executive support was given by Dave Fink (VP of R&D) and Mickey Steinburg. Innovative art direction was provided by Philip Freer, and later Gary Daines (who was mostly responsible for the hand-painted cel look of the *Aladdin* world). Composer and sound designer Aarin Richard brought the world to life with music and, perhaps more importantly, discovered the ability to combine the different types of sounds. The software technical lead for the project was Imagineer Scott Watson, who did most of the computer programming, with valuable help from the Silicon Graphics Performer™ team. Innovative hardware design came from Eric Haseltine, and Jesse Schell added the game elements to the Disneyland version.

The project was directed by George Scripner up to its presentation at EPCOT. During the EPCOT and Disneyland presentations, direction was by Bob Taylor. Taylor had considerable influence on the content that was shown to the public. This list is in no way complete. From 3D animators, to actors, to writers, many other people were involved with the creation of this application.

Many choices made by the designers creating this experience have been sprinkled throughout this section. But, because we consider content creation to be the most important element in this type of experience, the biggest design choices made were those involving the understanding and creation of intuitive ways for content designers to create the virtual world.

The visual aspects of the world were created using a combination of in-house software and *MultiGen 3D* modeling products. Character choreography was hand animated by in-house animators.

Sounds were recorded in locations similar to the virtual experience, for example, a marketplace. Musical scores were generated using traditional film-scoring techniques. Traditional sound effects techniques were also used for sounds associated with events, etc.

Narrative design was honed from being totally open ended to a mission game in order to match the time constraints of the venue. The open-ended approach was envisioned to be a world filled with interesting characters with which the guest could interact, generating a personalized story for each guests' experience. In the mission game approach, the overall plot is basically the same for each guest, although the order and timing of events may vary, and not all the guests' experiences will end with the scripted, victorious conclusion.

DESIGN CHOICE

Interactive design of the content is *very* important. The SAL (Story Animation Language) was developed specifically for this purpose. Artists (i.e., content developers) need the ability to quickly sketch out their ideas and see the results. If an artist has to wait too long between iterations, they will use a different tool or technique.

```
(spawn-task (spin-around guest clouds) `spin) (define (kill-when-stuck test-volume
max-time task) (defineguest-stuck0.0) (task-while (in-volume test-
volume guest)) (add!guest-stucktime-dt) (when(>max-timeguest-stuck) (rm-task
task-name) (kill)) )
(spawn-task (kill-when-stuck room-volume 20.0 `spin))
```

FIGURE 9-5

SAL code excerpt of a routine to help free a guest caught in a corner. (Watson)

The ability to revise scripts with rapid iterations is important to the creation of noninteractive scripts, and the design team felt this was also important in interactive narrative. Prior to the development of SAL, changes in the system required program compilation times of 4 to 5 hours, which were frequently done overnight. Thus iterations were basically slowed to one per day. If a compile failed, another half or full day was added to the cycle time.

The development of SAL allowed the team to iterate more frequently, making it easier for them to test new ideas. With SAL, a director or writer working on a project can make suggestions such as "What if you move this object over here?" "What if you change the timing on this event?" or "What if you make this object a little larger?" and see the results within a minute or two.

SAL (built on the Scheme computer language) was designed to optimize particular goals of the experience development:

- Building characters
- Creating animation of characters
- Interpreting commands provided by artists
- Synchronizing sound events with actions in the world

The SAL language was designed to be usable by animation and story artists. It allowed the artists a more direct means of expressing their ideas of the experience. Having a direct conduit helps to reduce the amount of human resources spent on entering the story, and reduce the pitfalls of misinterpretation that can occur when the writer communicates with the computer via someone else (who is more technically savvy) rather than communicating directly with the computer.

From their experience with feature films and theme park attractions, the developers knew the importance of audio in mentally immersing an audience into an experience. Using the layering technique described earlier, the main score can easily be combined with spatialized 3D sounds.

Buildings and objects in the world were impenetrable, so collision detection had to be calculated between the carpet and the walls and other objects in the world. They found this to be one of the bigger problems of the application development.[8]

In future projects, for performance and better flight, they will probably "cheat" by creating a second database specifically for determining collision events. This database would be in addition to the database for rendering the world.

With fully computed collision detection, people would sometimes get stuck in corners (e.g., between a staircase and a wall), causing them to bounce back and forth. This rapid bouncing results in the visual impression that the world is vibrating. A separate collision detection database could also be used to prevent flight into such corners and avoid the problem altogether.

Conclusion

This application was developed to explore the use of a new medium. Therefore, the developers were extremely interested in using their work to experiment with the features of the medium and analyze it. Trials were done with actual theme park attendees. Over the course of the EPCOT display,

[8] In fact, collision detection is a common difficulty for most projects that simulate interactive 3D worlds.

over 45,000 guests were selected to immersively experience *Aladdin* [Pausch et al.].

Each guest filled out a questionnaire containing questions regarding demographic information, what they liked and disliked the most about the experience, how it made them feel, and how much discomfort they felt. For some, this information was correlated to data gathered during the flight of the guest (their location and head orientation were stored several times a second). This logged data could then be used to analyze where guests flew and how much they moved their head, etc. [Pausch et al.]

Many results of these experiments have been published in *Disney's* Aladdin*: First Steps Toward Storytelling in Virtual Reality* [Pausch et al.]. Highlights include the relatively insignificant differences in the survey responses between male and female guests, a characteristic that also held for different age groups, which suggests that VR therefore has a broad appeal.[9]

It was hoped that analysis of the logged flight path data would show flight patterns and points of interest. They found, though, that people flew everywhere with no emergent patterns [Pausch et al.].

With such a long-running experiment, the designers were able to test many different iterations of the world, as well as many different system arrangements. Of particular interest was the correlation between the use of vestibular (motion) feedback and immersion (and nausea). The results suggest that when the motion base provided pitch movement that corresponded with the visual experience, guests gave a higher rating to the experience. However, there was also a higher report of nausea. To reduce the amount of guest nausea, they ultimately decided not to use the motion feedback.

The major accomplishment of the EPCOT *Aladdin* experience was the number and variety of people to whom it provided an immersive experience (over 45,000 from the general public). In addition to measuring some aspects of the guests' experience, it allowed the content creators to try many new ideas in this new medium. It also allowed people to enter a world they first experienced in a linear medium: motion pictures.

[9]Significantly, the questionnaires were designed in the hope of discovering differences in population segments.

The goal of this application was to experiment with and learn about virtual reality. The creators certainly met this goal, trying many different forms of navigation, display technology, and storytelling. Determining whether VR is a viable medium for a theme park, however, is a separate question that is still open. The fact that Disney chose to include VR experiences in their DisneyQuest venues suggests that VR may find a more permanent home in theme park and LBE environments.

It seems as though this application was successful at immersing the guests in the world of Agrabah. Survey respondents may have indicated this simply because it was their first use of virtual reality; however, a lot of effort was put into making the experience as immersive as possible within the constraints of the venue.

As we've indicated throughout this book, content is key, and having a good story with a familiar world certainly helps make the content more immersive. However, another important ingredient is the use of audio, especially audio that includes spatialized sound cues, sonically placing the user within a scene. Visually important was the fact that the world was an accurate representation of that provided by the Aladdin motion picture despite the fact that it was cartoon-like. In fact, choosing an animated world may have made it easier to immerse people. Pausch and colleagues distinguish between "photorealism" and "believability" [Pausch et.al., 1996]. They state that, in this case at least, the cartoon nature of the world increased the sense of "being there."

The designers made many discoveries in the course of creating this experience. Of the more significant discoveries was the ability to mix and layer different types of sounds. That is, sounds spatialized to a specific location in the world with generic 3D sounds, on top of a stereo background track. They also found that sound associated to (and spatially rendered with) an object adds to the sense of object permanence in a virtual world. Characters also seemed more believable if they made eye contact with the participant.

Many features of storytelling were uncovered in this experiment. One important feature is that people want something to do. They want a goal. Initially, people are willing to spend some amount of time exploring the world, but the smaller the world, the quicker people will lose interest without some direction. On the other hand, after the transition from the real world, participants need to have enough time to get their bearings and become familiar with the world in which they are immersed before they can set out with any sense of purpose. The length of time a participant may spend in the virtual world (as dictated by the venue) affects how open or directed the plot must be.

The major implementation discovery was that a separate database to handle collision detection between the participant and the objects (buildings, etc.) in the world would greatly simplify the task of keeping users from getting stuck in corners. At the same time, this would reduce the amount of computation needed for doing collision detection.

As mentioned, the *Aladdin* attraction was extended beyond the original test system, and installed in the Disneyland Starcade in 1996. This was also a temporary arrangement, and it is no longer available to the public. However, the DisneyQuest LBE commissioned the Imagineering VR Studio to produce a new *Aladdin* experience, as well as new VR experiences based on *Hercules* and *Pirates of the Caribbean*. Other VR experiences were installed at the LBEs, including *Ride the Comix* and a prehistoric river raft ride.

The developers are excited by the idea of giving the storyteller the ability to communicate directly with the senses. However, they feel that there are still some difficulties that currently hinder its use in the theme park venue. While the performance/price and availability are low, public VR venues often are more ride than story. Once the length of time available for participants to remain involved with the experience is extended, the ability for the storytellers to present a good interactive story in a 3D virtual world increases tremendously. As developer Gary Daines says: "If you don't have a goal, there's nothing to do—no story. This is true also for movies." After *Aladdin* was removed from the Disneyland Starcade venue, the immediate future for VR within Disney became in doubt. However, low-cost technology, and public acceptance of VR are becoming more widespread, so Disney continues to investigate potential VR projects for public entertainment.

As with any form of human communication, the message being transmitted is more important than the technology used for the exchange. Jon Snoddy drove this idea home when asked about the fidelity of the equipment used in the Aladdin experiences: "Content is the most important element (what you do), not the lens through which you perceive what you do."

CASE STUDY 9.2: VIRTUALITY GAME SYSTEMS

Many people who have experienced VR probably had their first exposure to virtual reality through VR games in an arcade, LBE, or theme park. In the future, home entertainment systems will probably be where more and more people will participate in virtual reality experiences. During the mid 1990s, this opportunity was primarily available on systems produced by Virtuality PLC. Virtuality was founded in 1987 and originally named W Industries. The company grew to become the world's largest supplier of LBE VR systems, with a 78% market share. Over 40 million people experienced VR games on one of their 1400 installed LBE systems.

The goal of Virtuality founder Dr. Jonathan Waldern and the Virtuality staff was to produce VR systems and experiences that were robust enough that they could be deployed in public spaces (and eventually homes) without requiring the venue operator or home user to be an expert in virtual reality or computer system integration. The company continued to innovate in new directions, establishing a professional VR development system—the Elysium series, developed with IBM, and a consumer prototype system developed for Atari. Unfortunately the Atari product never made it to market due to the demise of the Atari Jaguar platform for which it was designed. They also formed an

FIGURE 9-6

Many Virtuality games immerse the participant in a head-mounted display and provide a handheld controller that the participant uses to interact with the virtual world. A ring provides stability and creates a boundary for the participant's physical motion.
Photograph by William Sherman

Advanced Applications Division, which specialized in creating custom hardware and software implementations of Virtuality technology for corporate and industrial clients.

In many respects, the company was a victim of its own success. The company's exceptional growth during the first half of the 1990s led its new management[10] to invest heavily in inventory in order to speed up production of its entertainment systems. This decision proved costly during the downturn in the arcadeindustry in 1996 and led to a decline ending in the Virtuality Group filing for credit protection under Chapter 11 of the bankruptcy code in February 1997. However, the Virtuality name lives on. Cybermind GmbH, a Virtuality customer, cybercafe chain, and software development company headquartered in Berlin, Germany, acquired the Virtuality Entertainment business and continues to sell and support the systems worldwide. Dr. Waldern also returned to acquire the brand and certain technology assets he had been working on at Virtuality before founding his new company in California: Retinal Displays Inc.

[10] Dr. Waldern stepped down as CEO in 1995 and moved to the United States to pursue new display research interests.

Application Description

Two basic arcade configurations are offered on which the games are played. Both use a head-mounted display, with the graphics and simulation performed on a PC system with special graphics accelerator cards. The difference between the platforms is whether the participant stands and holds a handheld prop (model 2000SU) or sits with a steering and throttle input control mounted on the platform (model 2000SD). The VR games were designed with a specific interface platform in mind, depending on the travel interface involved.

In the stand-up system, the participant stands on a circular platform that also houses the computing equipment. A rail encircles the participant about waist high. This gives a limit to the distance the participant can physically walk, as well as something they can lean against or hold to maintain stability. The handheld prop is a 6-DOF tracked unit used for user input control (such as aiming and shooting, or activating something). Some games, such as the *Winchester Trap Master,* use specialized handheld props crafted to give a more realistic interface.

In the sit-down version, participants sit in a plastic pod with a bucket seat and a spoiler, giving it a carlike appearance. The armrests house the controllers that the participant uses to interact with the game.

The virtual world experiences appear as three-dimensional cartoon-like environments. The participant sees an avatar of the controller unit when appropriate to the game. For example, a rifle, a hand holding a pistol, or even the front of their vehicle. In multiplayer experiences, players see each other either as a human (standup platforms) or their vehicle (sit-down platforms).

As early as 1990, Virtuality realized that content was a crucial factor in how well VR arcade systems would be able to sustain market share and live beyond the initial "gee whiz, isn't VR cool" phase. Consequently, they hired someone with a background in film/TV production to help steer game development, Mike Adams. Recognizing that the content criteria for the home and traditional arcade game markets are significantly different, Adams realized how arcade games must be easy to learn and use, and provide a quick adrenaline rush worthy of the high cost per

FIGURE 9-7

Two basic platforms were configured for a wide variety of games. (Top left and bottom) Some games, such as Trap Master*, use special input devices such as the compressed air recoiling shotgun prop (top right).*

Images courtesy Virtuality PLC

game typically associated with these expensive systems. Games developed for home systems such as the Nintendo GameCube, Sony PlayStation 2, Microsoft Xbox can, or a home computer can be designed as larger worlds that can be explored in depth over a period of hours, days, or weeks.

Thus the bulk of Virtuality games were action oriented, and moved at a rapid pace for a rather brief period of play. As VR home units become available, we can expect to see games that are more thought provoking and/or exploratory in nature. Increased opportunities for exploration and investigation give more depth to the possible content, but would take longer to play than is economically feasible in an arcade.

According to Adams, then vice president of the Virtuality Advanced Applications Division, content in the VR medium is manifested more as a location to explore than as a linear

FIGURE 9-8
Screen shot of a Dactyl Nightmare *game showing the time remaining at the top and the opponent's avatar in the middle.*
Image courtesy Virtuality PLC

story line. The story can be embedded in clues supplied in various nooks and crannies of the virtual world. Thus, the story unfolds in different ways depending on how the participants go about investigating the area.

Because the Virtuality systems were designed as platforms capable of running a variety of games, we must describe a selection of these games to give a sense of what was accomplished and motivate future design ideas. The following is a brief summary of several of the more popular Virtuality game experiences.

Dactyl Nightmare

This is the game that started it all; the first game Virtuality made available in public venues. It quickly became somewhat of a classic among VR game enthusiasts. Typically two (the minimum), but up to four, players can compete in a world consisting of five multilevel platforms. Each person appears in the world as a humanoid avatar holding a weapon. The goal is for each participant to shoot the others as rapidly as they can while avoiding being shot. This game uses the stand-up platform treating the handheld controller as the firing weapon.

The players can travel freely in the small world by pressing a button to move in the desired direction. The game designers chose to use a combination of gaze direction and the vector between their body and their weapon as the direction in which players would travel. Initially they implemented a basic gaze-directed method of travel, but during focus tests with teenage subjects, they found that the new method provided a good balance that better met their expectations.

An on-screen display shows the score of each player and the time remaining. Each player's score is simply how many times they have successfully shot another player. The person with the most points when the time runs out wins. To add interest to the game, a pterodactyl flies overhead. In addition to trying to avoid being hit by the other players, a player must also avoid the pterodactyl. Otherwise, it might swoop down, grab them in its claws, take them 200 to 300 feet above the play area, and drop them to their death, reducing their score by 1 point.

A sequel, *Dactyl Nightmare 2,* was produced a few years later that moved to a world of tunnels with the added goal of collecting pterodactyl eggs, in addition to preventing

FIGURE 9-9
Screen shot of Dactyl Nightmare 2 *showing one player adding an egg to his stack.*
Image courtesy Virtuality PLC

FIGURE 9-10
Screen shot of Zone Hunter, *showing a view from behind the player's avatar as would be viewed by people observing the game on a monitor.*
Image courtesy Virtuality PLC

opponents from doing the same. Opponents are hindered in their collecting by being shot or by being captured by the swooping pterodactyl. By changing the primary goal of the experience to collecting eggs, the subsequent game could now be played solo.

Zone Hunter

Zone Hunter, the second game in the Virtuality lineup is an adrenaline-based game that can be played in a single- or dual- player mode. The scenario of *Zone Hunter* is that the players are to save the earth from a mass invasion.

FIGURE 9-11

Screen shot of Buggy Ball *showing multiple cars battling for control of the ball.*
Image courtesy Virtuality PLC

Requiring a minimum of two players was found to hinder the ability to attract customers, because it would require two people interested in playing the same game. The restriction of requiring multiple players in the original *Dactyl Nightmare* was removed in Virtuality's second release for the arcade systems.

This game was also designed for the stand-up platforms. Players travel through an alien spaceship as though riding on a conveyor belt (the ride-along method of travel). The participants can look and shoot behind them, but their motion is always forward. This is a high-speed, high-adrenaline, "shoot everything in sight" game. When more than one player is participating, the players ride side by side on the virtual conveyor belt. There are several rounds to this game, and a virtual commander speaks instructions at each new level. To proceed to the next level of difficulty, players must achieve a minimum score in the preceding level.

Buggy-Ball

Buggy Ball is a sit-down game for up to four players. It was created to provide a family-oriented, nonviolent alternative to most of the other arcade content available. Thus, it is a good choice for those who prefer a game without guns and a focus on shooting things. In this game, the participants find themselves in a wok-shaped arena, driving dune buggy–like vehicles. Points are scored by forcing a large ball out of the arena. The ball is moved by driving up to it and pushing it with the vehicle. Competitors try to steal the ball from whoever has control.

Each user can choose the type of vehicle they will drive, with each providing a different set of capabilities. The four choices were a four-wheel-drive jeep, a dune buggy, a bulldozer, and a "mad" police car. Also, the overall game difficulty could be adjusted to one of four levels: easy, medium, hard, and crazy.

Pac-Man VR

Pac-Man VR is another family-oriented game for people who wish to have a less violent game. *Pac-Man VR* is an immersive, three-dimensional version of the classic two-dimensional Pac-Man made famous by NAMCO in 1980. *Pac-Man VR* is interesting in that it is a transposition of a game that was authored for a two-dimensional video game medium into a three-dimensional game in the medium of VR. Each of the Pac-Man characters is rendered in three dimensions and the game is played in a three-dimensional world. Up to four participants can play simultaneously, and each participant takes the role of a Pac-Man, seeing the world from Pac-Man's first-person point of view. The goal (as in traditional Pac-Man) is to traverse the maze, gobble

FIGURE 9-12
Screen shots of Pac-Man VR *showing the items that might be encountered in the corridors.*
Images courtesy Virtuality PLC

FIGURE 9-13
Screen shot of Trap Master.
Image courtesy Virtuality PLC

pills, and work, either in collaboration or competition with the other players, to avoid the ghosts. Another classic early arcade game translated into the medium of VR by Virtuality is Atari's *Missile Command*.

Trap Master

In 1995, Virtuality released *Trap Master*. Commissioned by Winchester™, *Trap Master* is for the sportsman who enjoys trap shooting or someone who would like to try the sport in a simulated environment. *Trap Master* uses the standup platform, but provides an added natural interface via a prop of a life-like shotgun (a close replica of a Winchester 101 shotgun). The prop uses CO_2 cartridges to provide haptic feedback via the Total Recoil™ system, which simulates the kickback felt when shooting an actual shotgun.

Participants in *Trap Master* find themselves in a simulated environment that recreates an actual setting, complete with mountains, lake, horses, and evergreens. The system employs voice activation and allows the participant to shout "Pull!" when they wish for the system to launch a clay target. If the participant does not shout "Pull!" within 5 seconds, the clay target is automatically launched. Points are earned by hitting the targets. The target shatters when hit.

There are four skill levels available, and a virtual shooting coach is available to novice shooters with tips on aim, leading, and wind judgment. A round consists of 25 targets. Twenty or more hits earn the participant a bonus of two extra targets launched in rapid succession. Since the target launcher in the trap house is constantly moving, no two games are exactly alike. This game is available in sporting club venues, such as gun clubs.

Also using the Total Recoil hardware, the related game *QuickShot Carnival* provides a shooting gallery with fun,

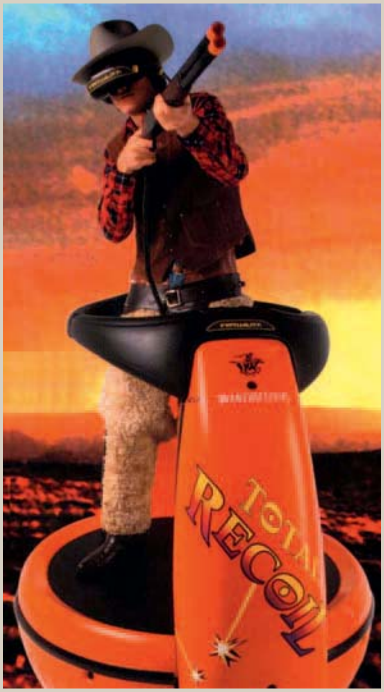

FIGURE 9-14

The Total Recoil version of the Virtuality hardware provides a realistic interface to the sport of virtual trap shooting.
Image courtesy Virtuality PLC

animated objects (such as leaping fish, marching soldiers, dive-bombing birds, shimmering tin cans, etc.) surrounding the participant. Participants shoot in rapid fire, and if they hit enough targets they get to advance to the next level. There are a total of eight levels, plus bonus scoring. The Total Recoil System earned Virtuality the "Best of Show" award at the arcade industry's annual IAAPA (International Association of Amusement Parks and Attractions) show in 1995.

Legend Quest

Probably the most ambitious game experience brought to the Virtuality system was *Legend Quest*. *Legend Quest* was a Dungeons and Dragons™–style game with multiple adventures that participants could engage in alone or with companions. *Legend Quest* was one of the few Virtuality experiences created by developers outside of the company, and was marketed in particular (LBE) venues specifically designed only for play of this game.

In many venues, monitors are available for spectators and people waiting in line to watch. Allowing onlookers to see what is happening inside the participants' HMDs helps to attract attention, as well as helps new players know what to expect and get a basic understanding of what to do in the experience. The venue operator helps to position and suit up players, and will often supply hints to novice players such as advising them to avoid only looking forward as most inexperienced players tend to do.

The arcade games would typically last under 5 minutes and generally cost on the order of $1 per minute. There have been Virtuality installations in many major cities in shopping malls, arcades, and other amusement venues worldwide. Additionally, arrangements have been made to have systems provided for special events and parties.

VR System

The Virtuality arcade systems all use high-durability, lightweight plastic HMD units designed by Virtuality. Named the *Visette,* the initial model weighed about 8 pounds and provided true stereo display from the dual graphics cards in the Amiga based Series 1000 systems. A later model, the *Visette2,* was lighter weight and higher resolution. Although the Visett2 includes dual displays (one for each eye), the Virtuality 2000 systems render only a single, monoscopic image displayed to both eyes. The designers felt that the loss of stereopsis was an acceptable trade-off for the lower cost of a single graphics card. The other 3D cues give a sense of immersion and three-dimensionality. Stereo headphones are integrated into the HMD for the sound display.

Tracking is implemented with the Polhemus electromagnetic tracking system. Both the head and handheld props are tracked in full 6-DOF, although not all games take advantage of this.

A single computer system handles the game simulation, the graphics and the sounds of each experience. In the (earlier) Series 1000 systems, a slightly modified Amiga computer system was used. Later, the 2000 Series systems introduced in 1994 used a PC-compatible system with graphics cards

FIGURE 9-15
A special HMD and tracker system were developed by Virtuality for interfacing with the Atari Jaguar home gaming console.
Images courtesy Virtuality PLC

designed specifically for the Virtuality system. At the time there were not many commercially available real-time graphics accelerator cards available for PC systems. A separate card handled audio computation and rendering.

In addition to the well-know arcade hardware, Virtuality also developed several other products or prototypes for the VR market. Some of these products include an inexpensive tracking system using infrared light called V-TRAK for the Atari Jaguar VR system. Unfortunately, this system never made it to market due to the demise of the Atari Jaguar game system. The Visette2 was sold independently from the arcade systems, allowing VR programmers to use the HMD on their workstations. The Elysium system (developed in partnership with IBM) was a complete integrated VR workstation with development software. Universities and research organizations were the primary customers for this system. Virtuality also researched and developed prototypes of products such as tactile feedback gloves.

Virtuality developed its own software library for interfacing with the arcade hardware and with the graphics accelerator hardware. With this software library, programmers had access to information about the user such as their full 6-DOF head position, hand-controller prop position for standup units, and driving controls for sit-down units. It was up to the programmer of the particular games as to how they would make use of the information. For example, not all games use the translational movement of the head to calculate the point of view, whereas in *Virtuality Boxing,* head location is very important and so had to be used to calculate the viewpoint. However, for applications such as *Zone Hunter,* where the crucial movement of the participant is through methods of travel other than physical locomotion, this feature was often excluded.

By providing a software library to handle the VR aspects of an experience, Virtuality hoped to attract other game design companies to write new games for (or port old games to) their arcade hardware platforms. Games developed by third parties included *Sphere,* a tank simulation for SD systems, and *Shoot for Loot,* a multilevel game show spoof by Gremlin Interactive in the UK. Unfortunately, the third-party developer initiative suffered from the typical business model in which the arcade business operates. Profit in the arcade game manufacturing business is made primarily from selling the hardware units.[11]

Since Virtuality provided the arcade hardware, it was not as advantageous for third parties to participate.

Development of games for the Virtuality arcade hardware took about 10 months, with the equivalent of four to five people at an overall cost of about $180,000. Each game had a producer, who would also be responsible for other projects, a lead programmer, an entry-level programmer, and varying amount of time from graphical artists to produce models and texture maps. A typical Sega arcade title might use 15 people over a year or two to produce.

Different content for different media

Aside from the differences in how software content is created for Virtuality arcade experiences, the differences in

[11] In the home video game business, most of the profit comes from sales of licensed software.

what content is appropriate for this medium as opposed to other forms of entertainment is also important. VR experiences differ from motion pictures in much the same way as other interactive computer-mediated entertainment in that motion pictures are presented in a preset, linear exposition. Computer-mediated experiences are not limited to a linear path through the storyline.

This difference leads to several implications for computer experiences. Where motion pictures can hide the unimplemented details of the virtual world being presented behind facades by using careful camera placement and other techniques, worlds presented via a virtual reality interface must fill in these details. The user's ability to choose a course of action in computer experiences means the world must be prepared to handle different possible paths, which greatly affects how the experience is plotted by the authors. In fact, the nature of who is the author is brought to the fore. While the audience bears some notion of authorship in a motion picture or play, this is much stronger in media whereby the audience can affect the storyline.

Another important implication of interactivity is the loss of timing. In motion pictures, the emotions of the audience can be controlled to a certain degree by how events are timed. In interactive media, timing generally cannot be precisely controlled. There are some specific cases where timing can be controlled in an interactive experience when some action must be performed in a set amount of time. Other possibilities are to affect timing of event by bringing agents to the user or having events occur to the user that force them to react.

Although some authors of interactive fiction experiences strive to create experiences with an existing story, while still giving the participant the sense of complete control of their actions, most computer experiences resolve this dilemma in one of two other ways: by reducing the actual freedom available to the player or by limiting the action to that of a particular scenes or battles. By limiting action to events that can be scored, such as a battle, the game can let the players act freely, and their actions affect the plot by determining one of two paths. Either the battle is won, and the story can continue, or the battle is lost, and the game is over. Thus, plot exposition (if there is any) is reserved for cut scenes, or a short story of what went before to bring things to their current state. Other experiences let players feel as though they are choosing their own course, but attempt to subtly direct their actions. Of course, if this is not done well, players will notice that their hands are tied, and their suspension of disbelief, or mimesis, will be broken.

When writing content for a new medium, such as VR, further restrictions are placed on what can reasonably be presented to the audience. For authors developing games for the Virtuality arcade systems, there was a broad set of criteria that affected the design of the games. Many of the criteria originate with the criteria of game design for any arcade game. First, one must assume they are writing for the first-time player. There must be clear and easy to follow rules. Games must give a 3-minute or so adrenaline rush and then give awards for accomplishing some goal. And, in head-based VR, when the user dons the HMD, they will be disoriented and will need time to adapt to the environment.

Probably the most constraining criterion is that the game must offer the quick adrenaline rush and chance of early accomplishment. Game designers for Virtuality arcade systems would have liked to have a game with lots of strategy and depth, but it is difficult if not impossible to create such games for this venue. Even home games such as *DOOM*™ or *Quake*™, which only offer a slight increase in control complexity over arcade games, rely on players to spend a lot of time before they begin to experience a sense of accomplishment. Richer, more complicated experiences generally require a delayed sense of satisfaction, preceded by hours of frustration. Lengthier VR arcade games cannot be developed expecting the players to pay for the hours of frustration experienced prior to accomplishment.

For Virtuality, *Zone Hunter* was the most popular title overall. This was due mostly to the fact that it was very simple and had a very low learning curve, and thus novice users could play the game without getting lost or stuck in corners. People already familiar with virtual reality preferred *Dactyl Nightmare* because it provided more controls, allowing the participants to roam freely.

Related Applications and Follow-on Work

Although Virtuality as we know it doesn't exist today, there has certainly been a progression in the creation of virtual reality and augmented reality games. One particularly

FIGURE 9-16
Sony's Eye of Judgment *game uses augmented reality technology.*

interesting augmented reality game targeted at the home video game market is Sony's *Eye of Judgment,* which runs on the PlayStation®3.

Eye of Judgment uses decks of cards with imagery as well as fiducial markers. A camera that attaches to the PlayStation is set up to point downward to where the cards are played. The system displays on the television monitor. When a player positions a card on the playing surface, the camera sees the fiducial marker and displays the appropriate (animated, three-dimensional) computer graphics on the screen. When another player places their card on the playing surface, the three-dimensional graphics related to that card are also displayed on the screen. When the cards are moved toward each other, the graphics then interact with each other.

Conclusion

Virtuality certainly set the bar for other public venue VR experiences by producing hardware that could withstand the rigors of regular public usage and providing a wide variety of game experiences. The main barrier to more widespread adoption was that the systems were expensive to purchase, operate, and play, and thus did not fit well in the established video arcade market.

At the time of introduction of such VR systems, video arcade operators typically purchased conventional game machines for about $5000 and charged $.50 per play (which might last several minutes). This is both cost-effective for the player and for the operator, who typically only needs to leave the machine sitting there and to periodically empty the money. However, the VR arcade systems generally require an attendant to help players get in and out of the equipment, plus control when the game begins and perhaps give instructions during play. For relatively short experiences, players paid about $1 per minute of play. This compared with the home option of purchasing a home game console at the initial retail price of roughly $300. If it takes the purchaser 70 hours to complete a home adventure-game experience, then their cost was $0.06 per minute, which assumes that the game and hardware have no further value.

Because arcade and other LBE environments offer a limited market for VR systems, Virtuality also made efforts to expand their market opportunities by creating an Advanced Applications Division. This division focused on the creation of custom VR applications and systems for corporate and industrial clients. Projects included marketing, training, and science museum presentations (what we might call "location-based education").

Their first marketing project was an experience brought to auto show trade floors demonstrating the Ford Galaxy.[12]

The experience consisted of boarding a virtual model of the car and being driven around by a chauffeur who explained about different features of the car. A research survey conducted at one exposition revealed that 77% of people wanted to do it again and 95% would recommend it to friends.

Another client was the Winchester gun manufacturing company. Winchester commissioned a clay target shooting experience to attract people to the sport at fairs and shows. This system proved so popular that Virtuality and Winchester agreed to market the system to LBE and arcade customers as well.

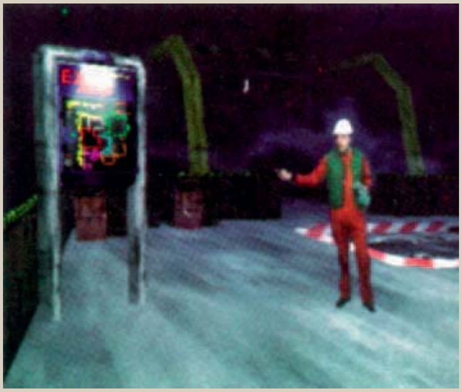

FIGURE 9-17
An instructor helps the user find the way out of a burning oil platform.
Image courtesy Virtuality PLC

In addition to direct product marketing, two training applications developed by Virtuality including a motorcycle driving trainer and an escape route safety trainer for oil platforms. The motorcycle trainer, developed for Kawasaki, includes a special platform with a motorcycle frame for the user to sit on. The platform includes the standard motorcycle control interface. The oil-rig escape trainer used video texture maps to display prerecorded agents who lead the user through the correct escape route.

The City of Nagoya, Japan, commissioned Virtuality to develop a VR experience in which children could learn about the interactions between humans and the environment. The result was a 48-person theater. The theater was designed to allow groups of 12 to work in four teams, with one member of each group wearing an HMD (the VR rider) and the others giving instructions to that user while looking at monitors for information about the virtual world. The goal of the experience is to solve an ecosystem problem. The team lands on the virtual world from space and discovers the world to be suffering from a slight case of decay. The team finds problems using its monitor, then instructs the VR rider on what life-forms to track down and photograph. The nonimmersed members of the team use the standard monitor to research the organism to learn how it relates to the ecosystem. The experience has different settings for different age groups. For the youngest ages, the system is mostly on autopilot and thus conveys the information as a story.

In 1996, Virtuality won a *CyberEdge Journal* CJ award for VR applications with its *RiverWorld* collaborative music application demonstrated at the SIGGRAPH '96 computer graphics conference. Produced by Mike Adams of Virtuality, the project itself was a collaborative project with the House of Blues, Philips, and Motorola Inc., with additional consulting by musician Thomas Dolby. Players at different sites could "play" specific instruments, and all participants could hear the combined sounds.

After Virtuality filed for Chapter 11 credit protection, the Entertainment Systems business was acquired by Cybermind, which continued to sell and support the systems worldwide. The Advanced Applications software development team went on to found a new company: Maelstrom Virtual Productions. Dr. Waldern himself founded Retinal Displays Inc., after leaving Virtuality under new management. Retinal Displays Inc. later developed new digital optical technologies of smaller, lighter, and higher-performance displays.

[12]The European name for the Windstar MPV.

FIGURE 9-18

A special interface platform developed for Kawasaki allows participants to learn how to ride a motorcycle. Details such as working mirrors enhance the simulation's usefulness as a training aid.

Images courtesy Virtuality PLC

The breakup of the Virtuality team does not, however, diminish their role in the history of the medium of virtual reality. Virtuality played an important role in introducing VR to the public in a way that could be directly experienced. It produced products and games that were durable and accessible. Virtuality founder Dr. Jonathan Waldern summarizes how improving technology will continue to drive the possibilities:

The first steps toward a mass market for VR were taken by Virtuality. Today, companies are developing critical optical technologies which will allow VR and many other high performance display oriented applications to become viable for the consumer market. The computer hardware and software technologies are already there. Evolving network technologies will soon enable real time multi-participant VR experiences that cross any geographic boundary.

CASE STUDY 9.3: VIRTUAL DIRECTOR

Executive Summary

Traditionally, computer graphics animators were required to create virtual camera moves using desktop tools such as a monoscopic screen and a mouse. In order to create camera moves and perform other actions in a three-dimensional world, the animator had to manipulate the camera in three dimensions using a two-dimensional display and a two-dimensional input device. Mapping the two-dimensional interface on, say, a standard desktop system into appropriate three-dimensional camera moves was a difficult task. The virtual director application allows the animator to move the camera in a three-dimensional display with a three-dimensional input device in a virtual environment.

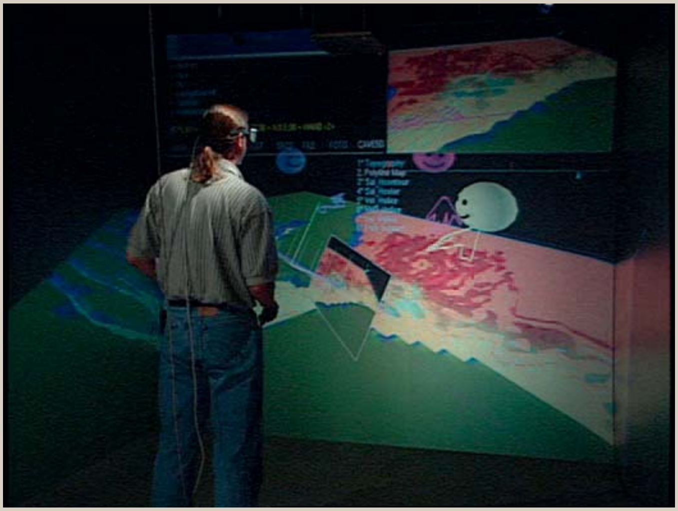

FIGURE 9-19
Bob Patterson, a computer graphics animator and co-developer of Virtual Director*, uses the virtual director application to collaborate on a scientific visualization project.*

In general, the output of the *VirDir* program is used to create traditional two-dimensional, linear movies. *VirDir* is particularly interesting for a variety of technological reasons. Because inputs needed to be very precise, as well as for other reasons, a speech interface was implemented. *VirDir* is also used in collaborative scenarios where animators, content experts, and other interested parties can collaborate at great distances geographically while sharing the same virtual world. The collaborators are connected via audio, as well as visually through the use of avatars.

Introduction

The *Virtual Director* application (*VirDir*) is a virtual reality tool that is a *navigation* and camera choreography tool for virtual environments as well as the desktop and acts as an aid in creating camera moves for computer graphics (CG) animations in multiple formats. The tool surrounds the user with a three-dimensional virtual world, and gives them a virtual camera with which to choreograph the camera's position in the virtual world. Voice input combined with gestures allows the director great flexibility in interactively creating camera moves. While *Virtual Director* can be used for entertainment purposes, the original application for the tool was to aid in the creation of movies of computer graphics for scientific understanding and communication. Indeed, the tool has been used as an aid creating scientific visualizations that are quite entertaining, or more specifically, "edutaining." The *Virtual Director* has been used in the production of computer animation for a variety of products, including a scientific visualization segment in the Oscar™-nominated IMAX film *Cosmic Voyage,* the PBS show *Mysteries of the Deep Space,* and animations of the ecological system of the Chesapeake Bay. It has found additional use in projects such as PBS's *Nova* HDTV special "Mapping the Universe," including simulations of cosmology.

The *Virtual Director* came about for a variety of reasons. The overriding goal was to create a tool that could be used in the motion picture production process. The primary focus of the *Virtual Director* has been to provide a variety of navigation tools to examine scene from all angles. A secondary goal was to create a tool for previsualizing camera moves for live-action shoots. Additional uses have also been envisioned, such as providing an interactive editing tool that could be used to create a rough cut of the product even before any filming takes place.

A catalyzing factor that helped to bring about the creation of the *Virtual Director* was that virtual reality equipment was available at the National Center for Supercomputing Applications (NCSA) where researchers also work on animations for scientific visualization. In particular, a *BOOM*™ head-based display was installed in 1991, followed by a *CAVE*™ projection-based VR display in 1994. Some of NCSA's computer graphics animators and scientific visualization producers were interested in how the VR equipment could be utilized as an aid in their work. After early test efforts were made using the *BOOM,* the current thrust of *VirDir* development began anew focused around a design for the *CAVE* and other projection-based VR displays.

A parallel goal for the system was to enable animators to do their work while minimizing the amount of typing required. This goal was precipitated by animator Bob Patterson, who was then working in computer graphics production at Industrial Light and Magic (ILM). Due to the severity of repetitive stress injuries (RSI) suffered from the amount of mouse motions done while creating computer animations for ILM, he returned to NCSA where he had previously worked. There he rejoined the push by Donna Cox to reestablish the efforts to create a "virtual director" driven by the need for a choreography tool for the making of *Cosmic Voyage.*

Thus, a design goal was that the *Virtual Director* would reduce the need for typing, mouse moves, and other discrete and repetitive physical inputs. Instead, voice input combined with fluid, whole-body movements are the primary method of input. Patterson looked to VR as an alternate way of working from the usual method of sitting in an office "typing, clicking, and dragging."

In the field of computer graphics, the scene is viewed through a virtual camera. This camera generally has parameters that correspond to a physical camera, such as focal distance, zoom, and a position relative to the objects in the scene. Before *Virtual Director,* camera paths for computer graphics animation were created on a desktop workstation running a tool such as *Alias/Wavefront* or *Alias/SoftImage.* Tools such as this have a very intense learning curve, and they take their input primarily from typed commands and mouse motions.

One of the more difficult aspects of using workstation tools is that they use a two-dimensional display to solve a three-dimensional problem. Hence, the user is forced to move the virtual camera in three-space via a two-dimensional interface. The moves were created by moving in one plane at a time. There had to be a better way.

For live-action shoots, camera moves are generally planned during production. While stand-ins are used instead of actors to help "block out" a shot, the crew and camera rentals eat into the budget, even if no film is shot. At $300,000.00 per day, time spent deciding how the camera should be placed and move in correspondence to the action can be very costly. Thus, it might prove prudent to use VR to previsualize the scene.

For live-action shots that will have computer-generated special effects added later, the placement of the camera is even more important. Having a preplanned camera path that is followed on the set helps the effects workers who do what is known as "match moving" to create a CG camera that follows the same path as the physical camera. On the flip side, having a virtual copy of the set and actions of the actors and props can be used on the set as CG puppets overlaid in the video playback of the shot. This allows the director to see a better approximation of what the image will look like after post-production.

Application Description

In contrast to creating experiences meant to be experienced in virtual reality, *Virtual Director* is used to exploit the three-dimensional, immersive qualities of VR to *produce* content meant for some other medium. Thus the VR experience of the CG animators is not what the final audience will see. The audience's view of the virtual world will be constrained to the scope of the virtual camera. In addition, the final rendering of the image may include many details only hinted at during the choreography work.

VirDir offers the computer animation producers a variety of tools to enable them to quickly create compelling camera moves for export to an animation system.

One thing that this allows is for directors to experiment with shots by themselves without relying on an animation technical director (TD) to be with them to support this activity. If directors care to, they can create the final camera moves themselves.

Representation of the Virtual World

The content in the virtual world of this application is dependent on the content of the virtual world of the production being created. Thus, if the *Virtual Director* is being applied to a film about star formations in the universe, the virtual world in this application contains stars. If the production is an animation for a television commercial, the virtual world is made up of characters and props related to the product. For previsualization of live-action scenes, the virtual world consists of virtual recreations of the live set, with CG representations of actors and props.

The virtual world of *VirDir* also includes elements of the user interface, the visualization of the camera path, plus avatar representations of other participants with whom the user is collaborating over the network.

Once camera moves have been choreographed, the move information is transferred to batch-animation software that renders the final results. The *Virtual Director* can be readily modified to create output suitable for many popular rendering and animation systems such as *Renderman*™, *Maya*™, and *SoftImage*™, as well as custom rendering software. For *Cosmic Voyage,* Loren Carpenter of Pixar developed a custom star particle renderer based on *Renderman*. Extra information can be stored about the camera path for renderers capable of calculating motion blur (e.g., *Renderman*).

The virtual world and interface elements are displayed in such a way as to clearly distinguish the interface controls from the scene. While the virtual world exists as a three-dimensional space that the user can travel through, most of the user interface elements are placed to appear on the surfaces of the VR display screens. This serves the multiple purposes of helping to distinguish between the virtual world and the user interface, as well as emphasizing the 2D nature of the final output and menu inputs. It also means that these elements can be more naturally viewed by the eyes because the focal distance and convergence cues will match. In fact, the stereo glasses are not even necessary to view 2D objects co-located with the screen. However, interface elements that are inherently 3D, such as the camera path representation, and avatars of collaborators are treated as part of the 3D virtual world.

The visual representational style of the animation world can be whatever the animators create. It can range from nearly photorealistic models of recognizable objects to abstract shapes, colors, and movements. The *Virtual Director* can be used to find a visual path through any form that can be represented by the computer. For example, for the IMAX movie *Cosmic Voyage,* stars and stellar gas make up the entirety of the animation world. Simple polygons and texture maps were used to represent the stars and gaseous material in the VR display.

The user interface of the virtual director has a very no-nonsense form. Menus, command inputs, and information about key frames and time values are given as text on the screen. Text is colored based on the type of information presented. For example, error messages are displayed in red to make them more prominent. The 2D "TV" view representation is simply a configurable rectangle on the screen with a rendered image inside. This TV can be formatted according to output style, such as HDTV, Imax, etc, and shows a rendering of the world rendered in 2D from the perspective of the computer graphics virtual camera.

Camera paths within the virtual world can be shown either as a series of points through the space, or these points can have axes added that show the direction that camera is looking, and a vector pointing toward the top of the view. Key frames are distinguished from the in-between positions of the camera by yellow squares. A dodecahedral wireframe highlights the current key frame. For the camera itself, the user can select different representations ranging from a simple forward/up axis pair to a "viewfinder" that shows an image of what will appear on the screen.

In the initial phases of the project, audio was not a high priority and was not implemented. Developers put forth several possibilities to improve the audio quality. One is to provide feedback to commands. The initial implementation was done via a text display. Another possibility would be to attach audio to objects in the virtual world scene. Also, sound could add the capability to do sonification in the environment, mapping sounds to various data parameters. This is important because *VirDir* is often used for producing scientific visualization.

Another potential use for audio is to enable the audio tracks from the production to be edited in the *CAVE* concurrently with the video production. This could be especially useful for editing three-dimensional audio in the scene.

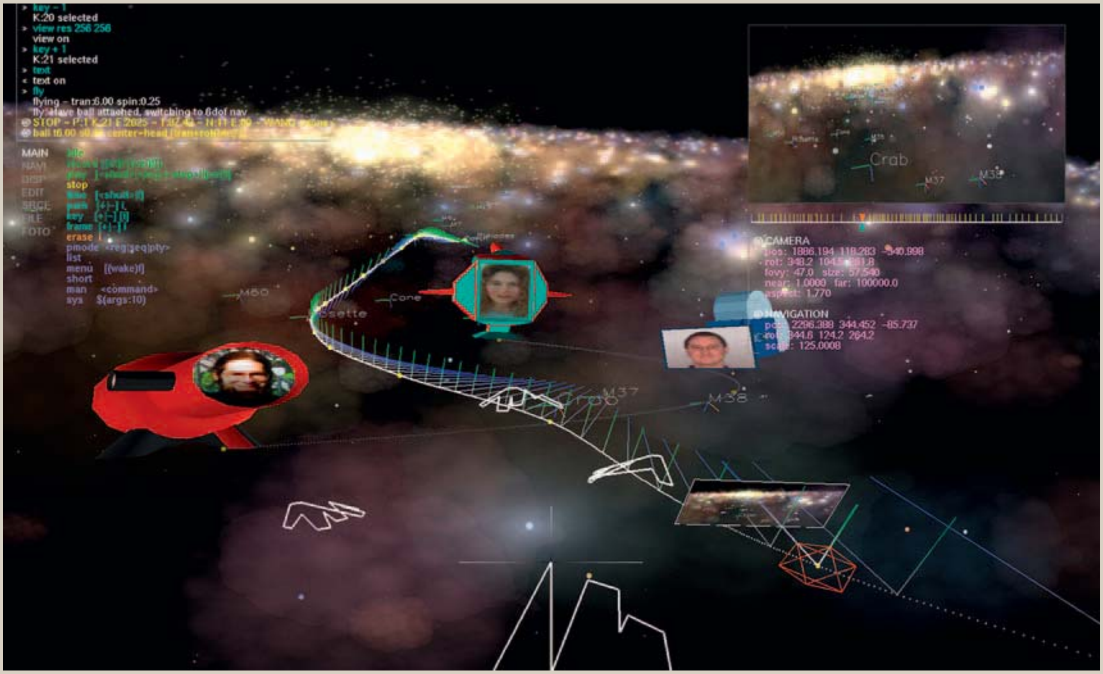

FIGURE 9-20

Many elements of the Virtual Director *user interface can be seen in this image. Collaborators are depicted by avatars that include their photograph, and an icon of their hand. The path of a computer graphics camera is shown as a line with axes that show the direction of the camera. Text information such as messages and commands can be seen in the upper left corner, while a two-dimensional image in the upper right corner shows a two-dimensional rendering of what the computer graphics camera sees.*

FIGURE 9-21

Two of the application developers collaborate in a CAVE *to fine-tune camera moves for a scientific visualization animation.*

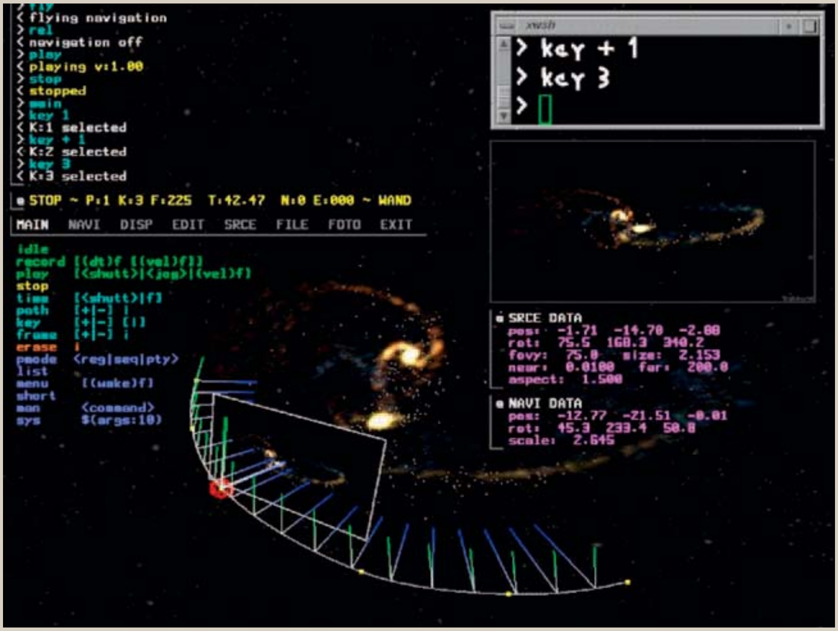

FIGURE 9-22

The Virtual Director *application allows speech input for its commands. In this image, the window in the upper right shows text translations of commands that were issued via the speech interface.*

Apart from holding the physical wand input device, there is no haptic feedback in this application. Originally the *Virtual Director* was displayed in a *BOOM,* which gave the feel of putting hands on a camera and directing it through the space. However, this was lost when the project was transferred to the *CAVE* system.

Since the *Virtual Director* is designed to create content for a medium that is limited to sights and sounds, the addition of other senses to the *VirDir* application itself must be beneficial toward this preeminent goal. The application designers and developers have not found a compelling reason to pursue the use of senses beyond mostly visual, and partially aural and haptic feedback.

Interaction with the Virtual World

The genre of this application is *iterative design*. It is often used by a single user, who begins by exploring the space and sketching out some camera moves, eventually refining the choreography to meet the narration goals of the animation. It can also be used as a *collaborative design tool* allowing multiple people to sketch path ideas, share interesting point-of-view paths and to comment on the ideas of others. The primary form of interaction with *Virtual Director* is free-form flying through the space, augmented with voice commands to govern the movement, and to adjust various features of the interface.

The narrative of this application is free exploration of the animation virtual world. The goal is to find interesting and insightful views and camera motions through the world. Participants can create camera moves that persist from session to session, allowing them to come back later and pick up where they left off.

Over the course of developing an animation, or live-action previsualization, the director will go through a sequence of phases that are all accommodated by the *Virtual Director.* At first, the director will simply want to explore the space of the world, flying around to determine where some

FIGURE 9-23

Virtual Director *has been used as a tool in the creation of many scientific visualization animations that have been seen by many people on television, in planetariums, and in IMAX™ movie theaters. This image shows a snapshot from the development of a scene illustrating colliding galaxies for the IMAX movie* Cosmic Voyage*.*

interesting and informative viewpoints are located. In the case of scientific visualizations, this will often be done in collaboration with the researcher that produced the data being visualized. The next phase is to "sketch" some preliminary camera paths and create rough animations from these paths to see how well they work. Once some good sketches have been produced, the director will probably create new paths based on the better sketches. Again these paths will be used to create test animations. If the results of the test animations are favorable, the director may go back into *VirDir* to fine tune a few key frames to create the final choreographed camera path.

In most cases, the scene for which the camera path is being created will itself have some form of narrative that evolves over time.

Because the goal of camera choreography is to help the viewer find their way through the space of the virtual world, it is vital that the "virtual director" be able to easily navigate the space themself. Thus, a variety of methods are available for the user to accomplish this task within the framework of the *Virtual Director* application. Using wand inputs in combination with voice commands, each of the different methods of travel are useful for different subtasks.

The most common method of travel is a style of *fly-through* control that allows the operator to simultaneously manipulate their location and orientation through the space. The particular method of flight control chosen is based on relative movement of the wand from the point where travel was activated. Translation is controlled by moving the wand fore and aft from the original position. Orientation is controlled by rotating the wand about its axes. Because users can begin by slowly moving the wand and gradually increase the offset from the "zero point," they can easily do moves that follow the traditional CG maneuvers of "ease-in" and "ease-out." The degree to which the particular magnitude of arm and wrist movements are translated to flight through the world is defined by the users, so they can select whether they want gross arm movements to be necessary to move at high speeds, or if only subtle physical movements will produce the same results. Voice command activates and deactivates wand-gesture input to the various functions.

As the user becomes skilled with the *Virtual Director,* they learn to perform several common physical maneuvers for different kinds of moves. An example of a move that is straightforward in *VirDir* is to choose an object and orbit about it while keeping the object in view of the camera. While this type of shot is not overly difficult to accomplish on a workstation screen using a mouse for a stationary object or point of interest, if the object itself is moving or there are other complicating factors, then the desktop method can quickly become tedious, requiring the user to move the camera a bit, then adjust the angle a bit, followed by moving the camera a bit, and again changing the angle a bit, ad nauseum. Such a camera move can be easily accomplished with *Virtual Director* by using a tool that automatically keeps the camera centered on a specific point of interest. However, while this is useful for a novice user, Patterson finds this mechanism too artistically constraining and instead chooses to maneuver the wand in three-dimensional space to achieve a satisfactory result. The *VirDir* point-of-interest tool allows camera-orbiting capability that allows orbiting around any point of interest, including camera paths while tracking some phenomenon. A second form of *fly-through* travel control is also available that is a little easier to use, though less flexible. This method of travel was added to allow novice users to explore the virtual world without requiring a lot of practice to become familiar with the controls. This can be important when a scientist collaborating with the director wants to point out an important visual element of the world, even though the scientist might not be a regular user of *VirDir.*

For some animations, the director may desire to affect the relationship between the scale of the world and the viewer. To this end, controls over changing (travelling through) scale are also provided. In life-like shots, the scale is generally set to be 1:1 with the real world. However, in many visualizations, such a scale would not serve the purpose of the animation, which is to allow the viewer to see the interesting features of the world.

The ability to adjust scale also provides for a method of travel we refer to as *scale-the-world* travel. In *Virtual Director,* scale-the-world travel is accomplished by scaling the user up relative to the size of the world about the point at which they are currently located. By them physically moving to another location in the apparently diminished world, and rescaling to the original size, the user is now located at the new location. In the case of *VirDir,* scaling is done about the location of the wand, so the user can move with a few simple movements of the wand. Scaling the user relative to the world also achieves the effect of allowing the user to make fine or gross location adjustments with the wand.

In addition to the fly-through modes of travel, the user can also *jump travel* to particular positions in the data. Voice commands allow users to select a particular (existing) key frame they would like to jump to, or they can jump to preset locations and times, or for particular types of scientific simulation data, to a region based on the resolution of the data.

Jumps to key frames move the user to the location of the frame without changing their orientation with respect to the world. Preset jumps can also be set to jump to a location while retaining the orientation, or can be set to orient the viewer to a particular view. Presets can also set the scale between the user and the data. For scientific datasets with varying degrees of resolution based on the data (referred to as Adaptive Mesh Refinement (AMR) simulations), the user jumps to the specified region at the appropriate scale for viewing the specified region.

Existing camera paths can also be used to control travel in a manner familiar to film and video editors, through *shuttle control.* Since each camera path denotes a path through space (and time), the user can select a path and then move along it in animation time[13] by twisting the wand with their wrist.

As a control mechanism through story space and time, *VirDir* must pay equal attention to moving through time as it relates to the virtual world. Just as in live-action motion pictures, slow-motion photography, time-lapse photography, and gimmicks such as a rapidly moving clock or calendar pages falling away are used to indicate that time is passing in the virtual world at a different rate than the physical world of the audience, the same is often true of animations. This is often the case with scientific visualizations. As the camera

[13] By "animation time" we mean that the camera moves along the frames of the path as though shuttling through the animation.

moves from frame to frame (advancing in "animation time"), the time in the virtual world might remain fixed, flow at a steady but faster or slower pace than reality, or even make discrete jumps in time.

In the case of *Cosmic Voyage*'s colliding galaxy segment, 25 million years are covered over the course of 5 seconds of the audience members' lives.

Because the worlds that the camera is moving through are often dynamic, a camera position is defined by where the camera is in space as well as exactly when in time. Thus, the application is dealing with time in three ways: data time, animation time, and real time. Part of the difficulty in maintaining acceptable results is that the various methods of keeping time must be unified in a harmonious fashion. The time-varying data does not necessarily vary at the same rate as the rate the frames in the animation are being displayed. For example, if there are 18 time steps per second in the dynamic dataset, yet the animation will be played at 30 frames per second, the application must reconcile what data time is to be displayed in each frame of the animation and, if necessary, create appropriate data by interpolation or other means.

Other than turning on and off menus and the "TV" display, the main objects within the *Virtual Director* that can be manipulated are the interface elements that exist within the virtual world: key frames, and whole camera paths. Additionally, the user can turn on/off the display of visualization elements, change isosurface values, create data slices, etc.

It is important to note that the aforementioned methods of travel through the space did not necessitate that camera paths were being created. In fact, users can fly the camera through the space without changing their own view of the world. For example the animator can move the virtual camera through the world much like a child might fly a toy airplane through the space around them without letting go of it. However, these methods of travel are a necessary part of camera path creation.

Directors can choose to create camera paths in a variety of ways. They can physically fly the handheld prop through the scene. They can discretely create keyframes and allow the application to create a spline path from key frame to key frame. The third option is to capture the path taken as they fly-through the world. The first choice listed (physically moving the wand through the scene) is the simplest, but also the least used method of creating a camera path. This is likely due to the fact that to accomplish this, the entire scene must be scaled to fit within the *CAVE*, and that situation doesn't allow for subtle control.

One common method of camera path creation is to find a few interesting views and connect them together by moving to each view, voicing the command to create a key frame at the current position, adjusting time, and then moving to the next view adjusting the key frame. The application developers and users state that this technique is very valuable, as long as the user maintains an awareness of their scale and the rate of passage of time. *Virtual Director* provides visual representations to aid that awareness.

Another common method of camera path creation is to maneuver to a good starting view and then activate a travel-recording mode such that either the position of the user's head or the center of the *CAVE* are used as input to the camera. If the user does not travel anywhere, then the camera will remain stationary (though time will pass in the simulated world). As the user flies through the world, the camera will follow along dropping key frames at regular intervals, creating a path that will be mimicked upon playback.

By creating multiple camera paths, camera path sequences can be created and edited. There are several possible sequence-editing constructs in the system, including:

- *Regular,* in which the camera stays on one path
- *Priority,* where the camera follows a master path, but can be overridden at any point in time by a higher priority move for brief parts
- *Sequential,* in which each camera path is executed in order, even if that means jumping back in time to begin the next sequence
- *Edit list management,* though not yet implemented, the future plans for *Virtual Director* include the capability of allowing any arbitrary sequence of paths and portions of paths to be strung together in a straightforward manner

The editing of camera paths is done by adjusting, adding, or deleting key frames on the path.

FIGURE 9-24

Many collaborators are visible in this image. These scientists are collaborating on their investigations of the Chesapeake Bay.

Each key frame contains information about the position in space and time of the camera within the simulated world. The actual camera path is a spline curve fitted to smoothly pass through all of the key frames. Fine-tuning a camera path is done by manipulating individual key frames.

Key frames are individually adjusted using menu input and wand movements to perform operations such as slight translations and/or rotations, gross movements, or deleting the key frame altogether. A key frame to be modified or deleted is selected by voice. Once selected, the key frame can be adjusted with the wand using either *relative* or *absolute* manipulation. Absolute manipulation causes the key frame to jump to the current position of the wand. When this is done to key frames that are a large distance from the viewer, the key frame will be moved considerably, and in the process is likely to terribly distort the camera path on which it resides. This problem is avoided by using the relative manipulation mode that only moves the key frame from its current position in the same direction and rotation as the wand is moved. Fine control is obtained by moving the key frame a fraction of the wand movement.

For some applications of *Virtual Director* it is possible for the camera choreographer to interact with objects in the scene of the simulated world, as opposed to only with the camera and key frames, for example, enabling a view of fish concentrations in a body of water for one segment of the animation. To allow this, *VirDir* uses hooks into the simulated world to manipulate the world. A submenu is provided under the normal menu selections, giving command access to the user for manipulating the world.

No agents are added to the virtual world by the *Virtual Director* application. While some of the worlds themselves may incorporate avatars that represent characters in the animation or live-action plan, these avatars do not interact with the director, and thus do not qualify.

The *Virtual Director* application includes a set of features to allow collaborators to work together in the virtual

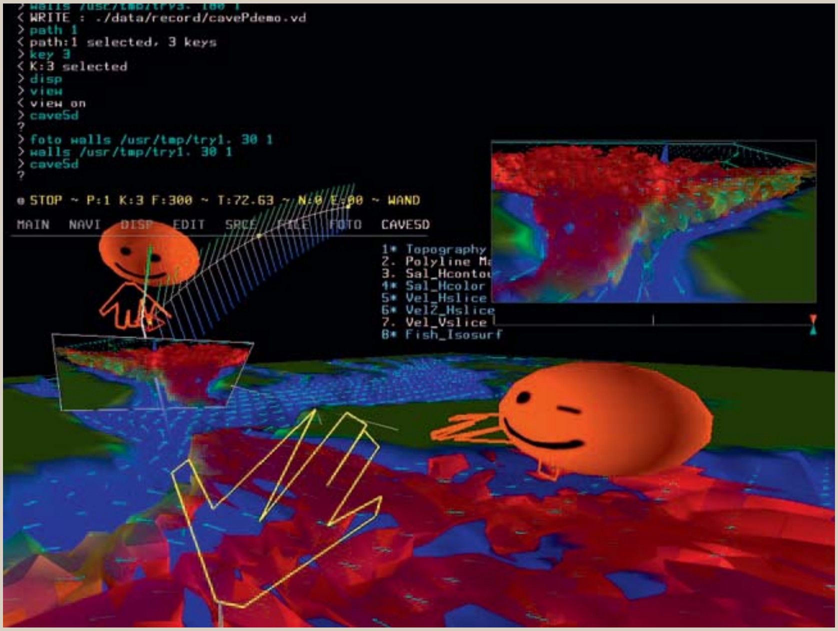

FIGURE 9-25

In this image, collaborators are depicted by spheres with faces rather than photographs. The participants can indicate facial expressions such as the winking eye on the avatar on the right.

space over vast distances. One such example of this was a collaboration between NCSA and Old Dominion University (ODU) in which the scientists in Virginia (Glen Wheless and Cathy Lascara), who were studying the ecosystem of the Chesapeake Bay, worked with the animation team in Illinois to jointly create an animation that shows the important scientific features of the simulated world.

Specifically, features added to support collaborative work include avatar representations of all the parties connected to a *VirDir* session, options to share choreography data (camera paths) between sites, as well as allowing users to broadcast commands to other sites for remote control.

Voice communication between sites is not supported directly within the *Virtual Director* application. Rather, voice is transmitted either by telephony communication links, Internet (e.g., M-bone) communication tools, or (when the collaborators are across the hall rather than across the country) via direct microphone-amplifier-speaker connections.

In general, the *Virtual Director* operates as a "no physical law" system, allowing the user to maneuver through the space unconstrained. The virtual world being explored in *VirDir* might operate under its own, separate world physics, however. Since the virtual director was built first and foremost for scientific visualization applications, most of the datasets displayed in the world are driven by a scientific simulation. Thus, the objects in those worlds adhere to the rules enforced by the simulation code. When it comes time to produce a visualization of the simulation however, the camera is not affected by gravity, high winds, galactic collisions, or otherwise.

However, in the case of preplanning a live-action shot, it would be desirable to have the virtual camera operate under the same physical laws in which the real camera will have to operate. Thus, for preplanning a live shoot the virtual camera should be able to be made to follow constraints imposed by cranes, tracks, and other real-world camera manipulation

FIGURE 9-26

The Virtual Director *Application has been used with many different hardware display configurations. Here, Donna Cox, a developer of* Virtual Director *uses the NCSA PowerWall as her means of entering a collaborative session.*

tools. Real-world production equipment simply can't move as freely as a virtual camera in a virtual world, and it is important to consider those constraints in the planning for a live shoot. Features to support a live-action mode are planned for a future version of the application.

Venue

The *Virtual Director* currently runs in the *CAVE* at NCSA and a few other research institutions. It also operates reasonably well on large wall and table VR displays. The developers would also like to see it in operation at movie production facilities, but this has not occurred yet.

Currently, the *Virtual Director* is designed for a single person operating the controls. However, being a *CAVE* application, it is easy to have a small group in the *CAVE* seeing the view as determined by the primary (tracked) user.

A typical session with the *Virtual Director* during the making of the *Cosmic Voyage* was 1.5 to 2 hours. Often the sessions extended to 3 hours, and sometimes became 6-hour marathon sessions immersed in the *Virtual Director.*

Any discomfort felt from using the *Virtual Director* for extended periods of time were attributed mostly to "issues of the contraptions," that is, pressure on the head from the glasses and tracker, rather than from other aspects of being immersed in a virtual environment. For relief and relaxation during the long sessions, Patterson enjoyed listening to music with the lights dimmed while he used the tool he and his collaborators had created. He felt the lighting and music contributed to creating a less tense, more creative environment.

VR System

The VR system supporting the *Virtual Director* is a typical *CAVE* environment consisting of the *CAVE* screens, head tracking with the Ascension Flock of Birds, and the typical tracked *CAVE* wand. The computing power is provided by a Silicon Graphics Onyx II system with Infinite Reality graphics hardware.

In addition to the normal *CAVE* system, voice input is provided by a small, head-set wireless microphone connected to a PC running *Dragon Dictate*™ voice recognition software. The voice software can be combined with software that allows the user to send text commands and control windowing and other system commands, enhancing the power of the voice interface.

The voice system in this particular application is speaker dependent and must be trained for the particular speaker. This makes the whole *Virtual Director* system unsuitable for random, walk-in users. Each user must train the voice system for their voice prior to using the *Virtual Director.* Each user can train the system to recognize aliases as well, so training the voice system is one way of creating macros and user-specific commands. These single commands can actually execute a string of *VirDir* commands. Another aspect of the training is that the user can allow the voice software to have many different voice commands map to the same activity. This prevents the user from getting tired of saying the same commands over and over.

Anecdotal evidence indicates that the voice system works correctly about 95% of the time as long as there are no other people present who are talking. Soft music playing in the background does not seem to interfere; however, human voices do. Thus, if there are kibitzers in the *CAVE,* their voices can confuse the voice-recognition system. As an aid in preventing this problem, one of the users (a male, Patterson) prefers a push-to-talk switch on the microphone. Another primary user of the system (a female, Cox) prefers instead to use commands such as "wake up" and "go to sleep" to activate and deactivate the voice system. Cox says she doesn't like "futzing around in the dark trying to find the switch." She notes that one possible reason for this preference is that her women's clothing does not have pockets onto which the switch can be clipped, whereas the male user, wearing jeans, does.

Application Implementation

The development of this application began in the early 1990s at the behest of Cox and Patterson. The first, experimental version utilized a FakeSpace *BOOM*™ device for display and tracking. The *BOOM* looked and handled somewhat like a camera and gave a camera-like feel to the operator.

In this version, there was no voice recognition. Instead, input from finger gestures were measured by a VPL data glove to issue four commands to the system. The four commands were data time forward, data time backward, data time stop, and record a camera move. The users found that using the glove as a gesture input device was difficult because users had to remove one of their hands from the typically two-handed task of moving the *BOOM* display. Also, it required calibration for each usage, and still only provided a handful of commands. Another problem with this version was that the *BOOM* offered only a "through the camera" view.

The user was not able to see any of the world beyond the camera as is possible in the *CAVE*. The original version was programmed mostly by Peter Fritzen, a computer graphics programmer visiting NCSA from the Fraunhaufer Computer Graphics Institute in Darmstadt, Germany. Some additional programming was done by University of Illinois undergraduate programmer Brygg Ulmer.

A new version was built from scratch by graduate student Marcus Thiebaux when a *CAVE* was installed at NCSA in 1994. The project team for this version consisted primarily of Cox and Patterson of NCSA as the designers of the *Virtual Director,* and Marcus Thiebaux from the Electronic Visualization Lab at the University of Illinois at Chicago as the primary programmer. Development of this version was handed off to a single other (but full-time) programmer (Stuart Levy) in 1997, when Thiebaux graduated.

The *Virtual Director* library uses *OpenGL* as the graphics interface, and uses the *CAVE* libraries to handle the tracking tasks, multiple-wall display, and stereo display. As a library, *VirDir* is in fact not a single application, but a suite of applications that use the same interface. New real-time rendering code is written for each application and linked to the *VirDir* library. In some cases, the entire rendering system is written from scratch (e.g., *The Cosmic Voyage*), while in other cases, existing code is merged with the *Virtual Director* as was the case for the Chesapeake Bay visualization.

Because the *CAVE* and other system hardware and software already were available, the primary cost in developing the *Virtual Director* was the time of the designers and programmer.

The *Virtual Director* system has been tested by (in fact, spearheaded by) the intended users, who were primarily computer animation artists. A variety of different interfaces have been tried. Some of them continue to be used, others are not. It is, however, significant to note that much of the early interface design was done by the programmer (also an artist) who worked alone to bring ideas discussed with the users to fruition. Many times, this resulted in the expectations of the users not being met. Other times, the expectations were surpassed. Thus, along the way, the overall design was the result of give and take on the part of the users and the programmers based on what was considered desirable versus what was considered feasible to program.

This project was developed at NCSA as a long-term development project. Often, in production studios, only a small amount of time is available to experiment with new techniques before one must go into full production mode. At NCSA, the designers have been able to experiment and develop the *Virtual Director* as an ongoing project, and investigate new ways of doing computer graphics animation production. Thus, major shifts in the design such as moving from the *BOOM* interface to the *CAVE* were not out of the question.

One of the biggest design changes was moving from the *BOOM* VR environment to the *CAVE*. There were several reasons for making this switch. One was a very pragmatic reason. The programmer (Thiebaux) was more familiar with programming in the *CAVE* environment. This, however, was not sufficient reason alone to justify the switch. A major reason was to gain the view of the world beyond the camera that the *BOOM* was not able to offer in an intuitive way. This was chiefly possible because the field of view in the *CAVE* was much wider than that in the *BOOM*. Also, the *BOOM* that was available at NCSA was an older unit with low resolution and a monochromatic display. In the *CAVE*, the user was able to "see" the computer graphics camera, whereas with the *BOOM*, the only view was through the camera. The *CAVE* that was available was much higher resolution, and color. The final reason for the switch was that the *BOOM* was somewhat limited in the possible motions the display head of the *BOOM* could move. For example, the *BOOM* cannot be rotated directly about the Z-axis. This was a move that was sometimes desired for virtual camera paths.

In the *CAVE* available at NCSA, the primary method of input was a wand with three buttons and a pressure joystick. Because the main objective was to create a path through space, the prototype version for the *CAVE* managed to accomplish this, and a simple key-frame editing tool using only the wand buttons. However, this offered only the "physically move the wand through space" method of path creation ("performance" mode) and provided only a fraction of the necessary functionality, and there were no more buttons available on the wand. To allow a more robust set of controls, a broader input mechanism was necessary.

For the next version, a system of menus was chosen as the best means to allow for all the options desired for controlling the animation camera. By this time, it had been decided that voice would be a crucial element for input, so the menu system was designed to operate off of text commands, that is, from the keyboard. Since the voice system operated by emulating keyboard input, commands can be entered either by voice or by actually hitting keys on the keyboard.

Using the text method of input allowed a straightforward means of integrating voice software. With this method, the voice software only needed to convert voice commands into ASCII text strings that are fed to the command line menu interface.

In addition to solving the health problems associated with constant, multiple button/key presses, voice input was found to have another important benefit. When wand buttons are pressed, the wand inevitably moves. This can interfere with the desired input. By using voice, an event can be triggered while the wand is held steady.

There is a downside to using voice as well. When it works properly, voice input has a small but nonzero amount of time required to utter the command and for the command to be recognized. This poses a problem for inputs that need to be instantaneous. The other problem with voice recognition is that, while improving, it does not provide 100% accuracy. Users of the *Virtual Director* have learned to work within these limitations.

Since interacting in the *Virtual Director* is not a means unto itself (as is true for many other virtual reality applications), a method of determining the quality of the resultant animation and, ultimately, a method of creating the final animation are required aspects of the system. To be able to judge what the animation will look like requires both a good representation of what the animation package will produce and good real-time display of the simulated world to see how the action will appear to the audience.

In order to see a decent real-time rendering while using the tool, a representative sample of the full dataset is used.

In the case of scientific data this might mean a fraction of the particles in the world are shown, or more crudely formed shapes. In recreating a real-world set, a combination of cruder forms and lower resolution texture maps help alleviate the real-time crunch. When the final rendering is done in batch mode, a high-quality renderer generates the frames outside of real time, using the full dataset.

Typically, the camera path data is transferred to an animation package that performs the rendering using the renderer of choice. Occasionally, the choreographer will want to see a better representation of the animation without waiting for a batch animation job. To provide this option, the *Virtual Director* can use the computer's graphics hardware's rendering ability to render the images, compact them into an animation file, and play the animation. The quality is not as good as typical animation packages, but this provides a result that is improved over the real-time rendering, and thus lets the director continue working without leaving the VR environment.

For real-time rendering, the frame rate in the *Virtual Director* can be improved by turning many of the display options off. A common use of this is to turn off all displays except the "TV view," which is enlarged to fill the screen. Doing this gives a reasonable play back of the camera move. The fact that turning off the 3D view of the world allows the animation to be displayed monoscopically also helps the frame rate.

Conclusion

There have been no formal studies on the ergonomic benefits or increased productivity resulting from using the *Virtual Director*. However, as they say, "the proof is in the pudding," and it has passed this one important test: It has been used, and continues to be used, in the production of professional computer animations. Two prominent jobs in which *VirDir* has been put through a significant work cycle are the creation of the *Cosmic Voyage* colliding galaxy animation sequences, and subsequently, to choreograph a segment for the PBS series *Mysteries of Deep Space*.

Thus far, the system has primarily been used by the team that developed the application. However, other researchers (such as the oceanography team at ODU) have also used *VirDir* for their own animation work, and other researchers have expressed interest.

However, the *Virtual Director* has not yet infiltrated the movie studios. *VirDir* would need to be more closely integrated with existing computer graphics technology in a cost-effective manner if studios are to embrace it as part of their production process.

The main accomplishments of *Virtual Director* have already been espoused: the creation of multiple and significant animations for scientific visualization.

Although the developers say they do not "lose themselves" in the virtual space, they do find it to be more satisfying to create camera moves when they are surrounded by the world they are animating. Patterson says: "It's like your data is suspended in front of you, and around you. You're in it and you can place the camera in absolute space as if you're holding it. The floor makes the *CAVE* different than other displays because you can see that space goes on below you. It's like flying around space in a free-form manner on an invisible platform." Cox notices that she feels "immersed until the tethers pull my hair."

Patterson points out that it has become totally natural to place the camera in a 3D space. In fact, he has "stopped doing camera moves at the desktop when [he] can go down to the *CAVE* and produce a variety of moves easily in less time."

Over the course of development, many design options have been tried and discarded, or tried, kept, and expanded on. Lessons learned by the development team range from whether VR was even usable, to comparisons between different display paradigms, to the benefits VR can provide over desktop interfaces.

The application designers were reluctant to use VR at first because of the lower-quality imagery that is produced by real-time rendering systems. They were unsure if the live rendering would produce results useful for judging an animation camera move.

The design team also discovered that the *CAVE* seemed to be the most suitable VR display paradigm for this particular problem. One of the positive aspects of using the *CAVE* as the display device for the *Virtual Director* is the fact that it provides a "view beyond the camera." Because the target output of the *Virtual Director* is a flat screen, the front wall in the *CAVE* is considered the most important, but the side walls and floor also contribute to the user's ability to see what is in the periphery, and beyond the domain of the camera. This is much like what a camera operator in the real world would experience; they are able to see the camera's view through its lens, but also the much wider real world outside the camera. Because the front view is the most important, other large-screen projection display devices such as large wall displays have also been found to be suitable, but they are less effective at showing parts of the world that are in the periphery.

Another positive feature of working in the *CAVE* is that it offers a break from sitting "stagnant in front of a radiating monitor clicking and dragging." That is, the typical mode of operation with this application in the *CAVE* is walking about and using the arms actively. This is a much more active, holistically healthy environment than working in standard animation environments.

Voice input is also a key contributor to the holistic nature of the application (this was expected). But voice was found to have other benefits as well. One of the key advantages of using the voice software is that the user does not need to take their eyes off the scene to interact with the system. This can aid in not losing one's train of thought, or having to interrupt a camera path capture in order to issue commands.

When the project began, the developers were interested in exploring VR, but they were not interested in being VR content developers. They were interested in using VR as a tool for creating high-fidelity images meant for display in other media such as film and video. To this end, they have largely been successful. But there is more that can be done, and so they continue to push *Virtual Director* in new directions.

One future development that has been mentioned is to expand the feature set of *VirDir,* increasing its suitability for preplanning live-action camera shots. They also plan to improve the editing capabilities, including the inclusion of edit list features.

The developers envision a scenario in a production facility where a computer graphics studio might have six ImmersaDesks or other relatively low-cost displays running *Virtual Director* where technical directors can schedule blocks of time to create moves, test models, and other activities. Using the collaborative capability of the system, a film director who might be working in another city could use a local VR display to view and discuss camera moves with the technical directors (TDs). In the future, the developers plan to expand the system to enable *Virtual Director* to animate more than just the camera. Any object in the world, such as lights or world entities could be animated. Additionally, through integration with other software packages such as was done with *CAVE5D,* they can manipulate the representations of the scientific data being displayed from within *Virtual Director.*